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Predictive latent states for extreme vortex gust–airfoil interactions

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Extreme vortex gusts drive a post-stall airfoil through transient wake reorganisations that the gust parameters alone do not determine. We investigate whether a reduced state can retain this wake dynamics, remain suitable for forward prediction and be estimated from sparse wall pressure. Direct numerical simulations of a NACA 0012 airfoil at $\alpha = 14^\circ$ and Re = 5000, subjected to Taylor vortices of varying strength, size and offset, compare predictive, reconstruction-trained and proper-orthogonal-decomposition states at matched dimension. A controlled supervision study shows that, within the predictive formulation considered, a lift target is necessary to avoid latent collapse, a near-body force-element target improves peak-load readability, and wake supervision is required to retain the wake enstrophy and circulation. Lift stays linearly accessible from four coefficients over the dimensions tested, whereas the spatial wake structure requires a larger distributed representation. Under a common direct multi-horizon forecaster the predictive states give the most accurate lift forecasts, while recursive prediction is limited by accumulated rollout error. A pressure-only ensemble estimator using eight wall taps tracks the lift through the impact and relaxation phases, reaching impact-phase $R^2 = 0.794$ in distribution and 0.837 for the held-out $|G| = 4$ cases with the final two-stage configuration. Wake enstrophy is not reliably recovered by the wall-sensor configurations considered. The results separate the roles of observable supervision, predictive training and sensing geometry in constructing reduced states for extreme aerodynamic transients.

Key words: vortex shedding, low-dimensional models, machine learning

1. Introduction

Small uncrewed aerial vehicles are increasingly expected to operate where the incoming flow is neither steady nor weakly disturbed. Urban canyons, mountainous terrain, coastal regions and the wakes of ships and buildings can expose a small wing to gusts whose length scale is comparable to the chord and whose velocity is of the order of the flight speed (Jones *et al.* 2022; Mohamed *et al.* 2023; Fukami & Taira 2023). In this regime a gust is not a small perturbation of an otherwise steady aerodynamic state: the strongest encounters reorganise separation, vortex shedding and wake recovery rather than merely modulate the instantaneous force, and the resulting transient loads bear directly on navigability, control authority and flight safety.

This regime has motivated a growing body of work on extreme aerodynamics (Taira 2026). Its canonical problem is the interaction between a discrete vortex gust and a wing, where the disturbance is strong enough to reorganise the near wake rather than merely modulate the force (Fukami *et al.* 2025), and its central empirical discovery is that these apparently complex encounters admit unexpectedly low-dimensional descriptions. Lift-augmented autoencoders reveal compact manifolds for extreme vortex-gust airfoil interactions (Fukami & Taira 2023), the same reduced description supports phase-amplitude analysis and data-driven transient lift attenuation (Fukami *et al.* 2024), and large-scale vortical structures show similar organisation across laminar and turbulent extreme-gust encounters (Odaka *et al.* 2026). The dynamical organisation of these flows is therefore simpler than the full field dimension suggests.

For deployment, however, compactness alone is not sufficient. A reduced description intended for gust mitigation must serve as a plant state: it must be advanced in time by a low-order forecast model and estimated from the sparse sensors a wing can carry (Loiseau *et al.* 2018). Surface pressure is central here, because it is one of the few high-bandwidth measurements available on an aerodynamic body, and recent gust studies use sparse pressure histories to infer unsteady loads and to bridge spatial pressure correlations with temporal flow memory (Zhong *et al.* 2023; Haughn *et al.* 2024; Chen *et al.* 2026), while Bayesian sequential estimation has begun to close the loop between pressure and low-order aerodynamic states, carrying uncertainty alongside the estimate (Mousavi & Eldredge 2025; Eldredge & Mousavi 2025). These advances sharpen a prior question: what reduced state should be estimated from the wall in the first place?

The two established routes to such a state pull in opposite directions. Nonlinear reconstruction buys compactness but not forecastability. A decoder fixes the latent only up to an arbitrary reparametrisation of the manifold, so the coordinates a reconstruction objective produces need not be the coordinates a time-advancing model can use, and controlled comparisons in wake flows report that autoencoder latents mix high and low frequencies in their coordinates and evolve irregularly, so that autoregressive predictors built on them diverge, whereas linear proper-orthogonal-decomposition predictors forecast more accurately and with a narrower spread precisely because the basis is linear, but pay for it in compactness, needing many more coordinates to represent the same flow structures (Solera-Rico *et al.* 2025). The linear basis, however, buys that forecastable dynamics at the price of the nonlinear wake content of the encounter: in the present flow the POD coefficients do not reach the wake observables at any dimension we tested. Compactness trades against forecastability, and nonlinearity against wake reach, and extreme gusts are the arena where the trade cannot be evaded, because there the linear escape route genuinely fails.

This tension motivates a predictive training objective: rather than compressing for reconstruction and hoping the latent is forecastable, one trains the encoder together with a latent predictor so the coordinates are shaped in the space the forward model will use. This is a joint-embedding predictive architecture (LeCun 2022), which learns an encoder and a predictor such that the present latent predicts a future latent rather than reconstructing the field, developed for images and video as non-generative prediction in representation space (Assran *et al.* 2023, 2025). Here it is a diagnostic instrument for the paper's central question: whether predictive training yields a nonlinear state at once more compact than the linear basis and more forecastable than a reconstruction-trained latent.

Training such a state on a forced wake exposes a structural requirement an autoencoder does not face: without an observable anchor the predictive latent collapses, and it is the scalar lift, not a high-dimensional field descriptor, that prevents it. This turns the observable heads from an assumption into a finding and connects the objective to the observable-augmented manifold learning of the autoencoder lineage (Fukami & Taira 2023, 2025), in which physical observables orient the latent towards physically relevant quantities. A healthy latent and a useful one are different achievements: the lift anchors the state, and a wake target loads it with the wake.

We examine whether a reduced representation of an extreme vortex-gust encounter can satisfy three requirements simultaneously: it must retain the wake variables that distinguish post-impact states, remain usable under forward prediction, and be recoverable from sparse wall pressure. Direct numerical simulations of a post-stall NACA 0012 airfoil at $\alpha = 14°$ and $\mathrm{Re} = 5000$, perturbed by Taylor vortex gusts spanning gust ratio, core diameter and offset, are used to compare predictive, reconstruction-trained and proper-orthogonal-decomposition states at matched dimension. Controlled supervision ablations identify which physical targets organise the latent representation; a common direct forecaster separates representation quality from operator choice, the encoder never seeing the gust parameters and the reconstruction baselines carrying the identical observable heads; and a pressure-only sequential estimator tests which parts of the state are observable from the wall. The results show that observable supervision determines the physical content of the state, multi-step predictive training improves its forward stability, and wall-pressure histories recover the load-bearing directions more reliably than the complete wake state. The strongest gusts expose the remaining limitation of a mid-plane representation and an under-dispersed estimator.

The remainder of the paper is organised as follows. Section 2 describes the flow configuration, data and endpoints; §3 the reduced states, the shared forecaster and the wall-pressure estimator; §4 the results, in the order of construction (§4.1), representation across dimension (§4.2), the physics the state holds (§4.3), forecastability (§4.4), wall observability (§4.5) and sequential estimation (§4.6); §5 discusses the mechanisms and their limits, and §6 concludes.

2. Flow configuration, data and endpoints

2.1. *Vortex gust-airfoil interaction*

We study a NACA 0012 airfoil at angle of attack $\alpha = 14°$ and chord Reynolds number $\mathrm{Re} = 5000$, perturbed by a discrete vortex gust modelled as a spanwise-oriented Taylor vortex released upstream of the leading edge. The data analysed here are our own direct numerical simulations of this configuration, computed with the GPU-enabled spectral-element solver SOD2D (Gasparino *et al.* 2024) with no subgrid-scale model, so that all scales of the separated wake are resolved at this Reynolds number. The solver is run in the low-Mach regime in which the flow is effectively incompressible. The configuration and its physical characterisation follow Fukami, Smith & Taira (2025), which is the reference for the physics, not the source of the data. The vortex has the azimuthal velocity profile

$$u_\theta(r) = u_{\theta,\max}\,\frac{r}{R}\,\exp\left[\tfrac{1}{2}\left(1-\frac{r^2}{R^2}\right)\right], \tag{2.1}$$

which peaks at the core radius $r = R$ (Taylor 1918), and is parametrised by three scalars: the signed gust ratio $G \equiv u_{\theta,\max}/u_\infty$, the core diameter $D \equiv 2R/c$, and the wall-normal offset Y of the vortex centre, the displacement normal to the chord (the laboratory coordinates are rotated by the angle of attack to define Y). The released-data case identifiers encode the solver's opposite sign convention, $s = -G$, so a case labelled `G+4.00` in the archive corresponds to $G = -4$ here. The profile and the (G, D, Y) sampling follow the configuration of Fukami *et al.* (2025). The training envelope spans $|G| \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$ with $G = 0$ as the undisturbed reference, $D \in \{0.5, 1.0, 1.5\}$ chords, and $Y \in [-0.4, +0.4]$; the value $|G| = 4$ is held out as an extrapolation boundary, for reasons that are physical and that we return to below. Convective time t is measured in chords from the vortex release, $t = 0$, while t_{imp} denotes the instant at which the vortex centre reaches the leading edge, near $t/c = 2$ for this configuration; all phase windows below are referred to t_{imp}.

Without a gust, the $\alpha = 14°$ wake at this Reynolds number is massively separated and sheds periodically; in dynamical terms the undisturbed flow is a limit cycle. A gust encounter is then a transient excursion from this limit cycle and back, and it is helpful to read it in stages (figure 1) (Smith *et al.* 2024; Fukami *et al.* 2025; Odaka *et al.* 2026). The vortex first induces a strong wall-normal vorticity flux at the leading edge, the airfoil experiences a sharp lift excursion as a leading-edge vortex (LEV) forms, the LEV rolls up and sheds together with a trailing-edge structure as the load reaches its extremum, and the wake then relaxes back toward the baseline shedding. The sign of G sets whether the first lift excursion is positive or negative, Y sets the side and strength of the impingement, D sets how much of the encounter is dominated by the vortex core, and the impact timing sets the phase of the baseline cycle at which the disturbance arrives. The five observables defined in §2.2 are quantitative summaries of exactly this sequence.

A consequence of the source characterisation (Fukami *et al.* 2025) is central to how we read the extrapolation case. For $|G| \leq 3$ the encounter is primarily two-dimensional, so a mid-plane slice of the spanwise vorticity carries the relevant large-scale dynamics; for $|G| \geq 4$ the interaction becomes genuinely three-dimensional, with fine-scale instabilities developing during impact. We confirm this from the full three-dimensional vorticity fields: the fraction of the spanwise-vorticity (ω_z) enstrophy not captured by its spanwise mean, the content a mid-plane slice discards, is roughly constant across the training range ($|G| \leq 3$, post-impact wake median ≈ 0.20) and rises about threefold at $|G| = 4$ (≈ 0.56, computed on the four boundary cases for which the full three-dimensional fields were retained). Because the encoder in this work consumes a single mid-plane vorticity field (§2.2), the held-out $|G| = 4$ case is not merely one parameter step beyond the training range; it is a regime in which the mid-plane representation omits a substantially larger fraction of the three-dimensional vorticity content, so that more of the dynamics is not captured by what the encoder sees. We therefore treat $|G| = 4$ as a physically more demanding extrapolation set in which a mid-plane state is incomplete, rather than as a claim of parametric generalisation.

2.2. *Data, fields, and observables*

Each simulation case is one long run partitioned into encounters of 120 cached frames at $\Delta t = 0.05\,t/c$, one encounter per gust release. The simulations use a spectral-element discretisation of polynomial order $p = 4$, and the fields are cached on a structured

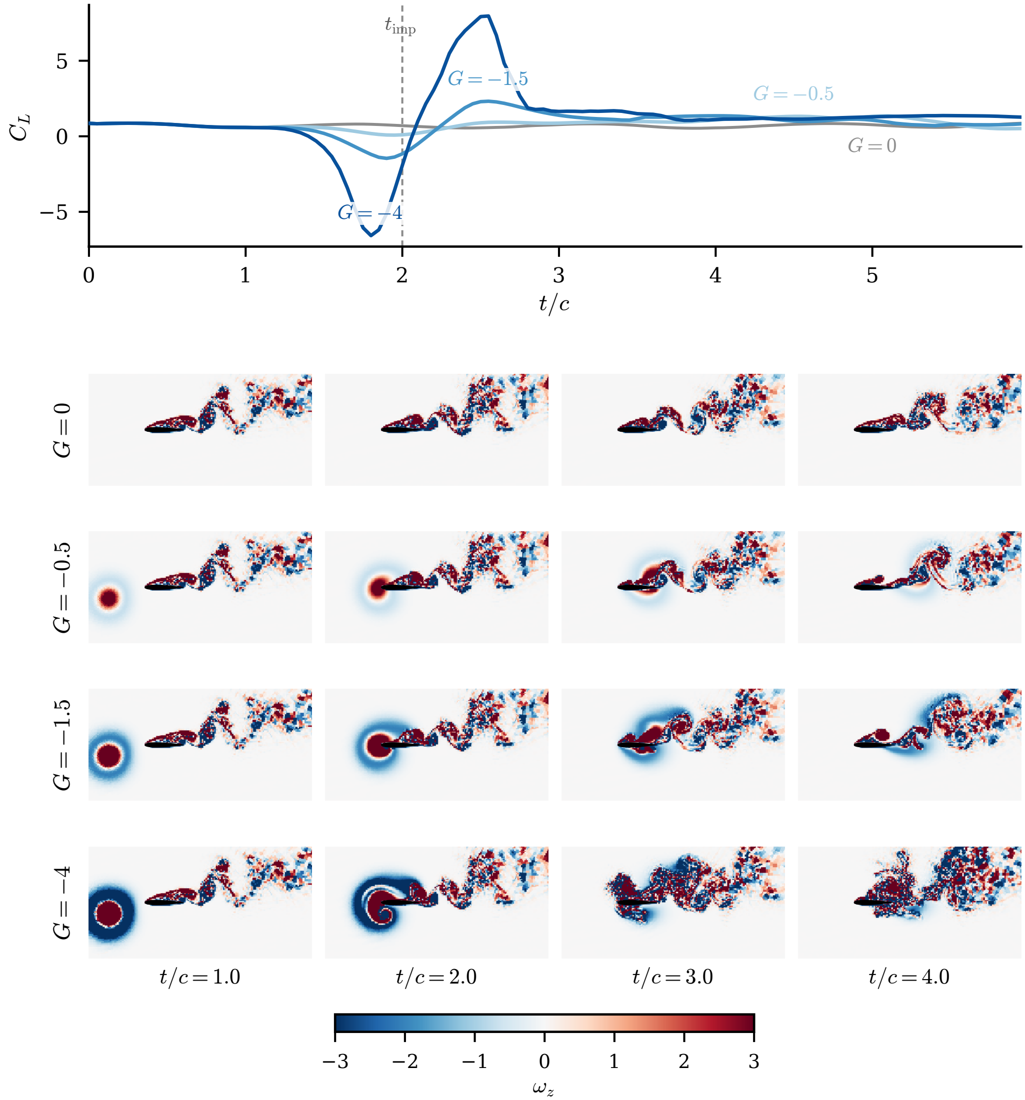


Figure 1. The vortex gust as a wake-reorganisation event: a gust-strength sweep at fixed core diameter $D = 0.5$ and offset $Y = -0.1$ (first encounter of each case). (a) Lift coefficient C_L against convective time, with $t/c = 0$ the release and $t_{\rm imp}$ (dashed) the leading-edge crossing near $t/c = 2$; the load excursion deepens and widens with $|G|$, the baseline ($G = 0$) in grey. (b) Mid-plane spanwise vorticity ω_z per case (rows $G = 0, -0.5, -1.5, -4$) at four times (columns $t/c = 1, 2, 3, 4$), raw units, colourbar clipped at ± 3 (intense cores saturate), airfoil filled. The strong gust reorganises the wake through impact while the weak cases barely perturb the shedding; $G = -4$ is the $|G| = 4$ extrapolation case, shown only for illustration. The release parameters alone do not fix where the wake sits afterwards, which is why the wake, not only C_L, is the endpoint a reduced state must carry.

grid of $192 \times 96 \times 32$ points spanning $x/c \in [-1.5, 4.5]$ and $y/c \in [-1.5, 1.5]$ with a spanwise extent of approximately one chord; this analysis subdomain matches the data-driven subdomain of the source characterisation (Fukami *et al.* 2025). The lift and drag coefficients are $C_L = F_L/(\frac{1}{2}\rho u_\infty^2 c)$ and $C_D = F_D/(\frac{1}{2}\rho u_\infty^2 c)$, obtained by integrating the pressure and viscous stresses over the airfoil surface. The remaining DNS solver-resolution details required for full reproducibility, the free-stream Mach number, the domain and spanwise extent, the element and solution-point counts, the near-wall spacing, the time step and CFL, the gust-release station, and the grid and time-step sensitivity, are listed as an author-fill checklist in table 1. The dataset comprises 102 cases partitioned into training, validation (the held-out last encounters of the training cases; archive name test_a), in-distribution test (test_b) and boundary test (test_c) sets; the archive names in parentheses label the corresponding records in the data release and appear only here. The in-distribution test set is stratified to span the gust strengths, the three core diameters, and the two source groups (figure 2); the boundary test set is the out-of-distribution $|G| = 4$ extrapolation set, sampled symmetrically in both signs of the gust (§2.2). These 102 cases are separated in preprocessing into 450 encounter windows of 120 frames each, one per gust release: 268 training, 100 validation, 42 in-distribution test and 40 boundary test encounters. The undisturbed reference case is retained in training. Behind these encounter counts are far fewer distinct cases: the 42 in-distribution test encounters come from only 10 held-out cases (6 interior and 4 boundary tier) and the 40 boundary test encounters from 8 cases, with most cases contributing four to six encounters that share a baseline shedding history. Encounters of the same case are therefore not independent, so encounter-level intervals overstate the effective sample size; the headline comparisons are reported additionally with a case-clustered bootstrap and a case-level paired test (Appendix A). The partition is fixed once and used identically by every model and baseline.

The $|G| = 4$ extrapolation boundary is sampled symmetrically by combining the periodic archive cases with four additionally released cases at the opposite sign of the gust. We verified the additional cases against the archive by a physical consistency check on the load rather than by trusting the file metadata: at matched core diameter and offset, the two signs of the $|G| = 4$ gust produce near mirror-image lift transients, one a positive-first excursion peaking near +8.3 and the other troughing near −7.6, as the sign of the gust ratio sets the sign of the first excursion (§2.1). This confirms that the two halves of the symmetric boundary set are physically consistent and that the sign-asymmetry analysis of the operating envelope (§4.6.2) is sound. Sampling the boundary at both signs is a genuine improvement over a one-sided $|G| = 4$ set: it lets the sign asymmetry be measured rather than assumed.

For each frame the cache stores the mid-plane spanwise vorticity $\omega_z(192, 96)$, the span-averaged wall pressure $p_{\text{wall}}(192)$, and the lift and drag coefficients C_L, C_D. The vorticity is normalised by a fixed pipeline: a spatial mask removes the solid and one adjacent cell layer to suppress the leading-edge finite-difference artefact, a per-encounter high-percentile clip caps spikes, and the field is scaled by three times the training standard deviation ($\sigma_{\text{train}} = 3.5396$) with no mean shift, preserving the vorticity antisymmetry. All representation and probe losses are computed in this normalised space; unnormalised values are used only for physical-units metrics and figures. The field-reconstruction similarity reported below uses the structural similarity index of Wang *et al.* (2004) with the standard constants $K_1 = 0.01$, $K_2 = 0.03$ and a dataset-dependent dynamic range $L = 8.487$, set to twice the global 99.9th percentile of the normalised target magnitude over the validation split so that the index is comparable across reruns. Appendix A repeats the primary wake probe under no amplitude clip and under a single training-pooled leak-free threshold (established on the previous-generation encoders; the pipeline is unchanged in this revision), and the family ordering is unchanged, so the per-encounter clip does not drive the result.

We evaluate forward closure against five physical observables that together span body-force and spatially distributed wake diagnostics. The lift and drag coefficients C_L and C_D are the body forces. The wake enstrophy and the positive and negative circulation are spatially distributed wake observables: they integrate (signed) vorticity over the wake region and are the most demanding to forecast, because they are sensitive to the redistribution of vorticity

Quantity	Value	Why a referee needs it
Free-stream Mach number (or incompressible-limit confirmation)	**[PENDING:]**	Confirms the low-Mach formulation is effectively incompressible at Re = 5000.
Computational domain and spanwise extent L_z/c	**[PENDING:]**	Bounds domain blockage and confirms the span resolves the three-dimensional wake.
Element and solution-point counts	**[PENDING:]**	Fixes the spatial resolution of the spectral-element discretisation.
Minimum wall-normal spacing $\Delta n_{\mathrm{min}}/c$ and wall units Δn^+	**[PENDING:]**	Shows the near-wall layer is resolved without a wall model.
Time step $\Delta t\, u_\infty/c$ and maximum CFL	**[PENDING:]**	Establishes the temporal resolution and stability of the integration.
Gust-release station x_0/c	**[PENDING:]**	Sets the upstream distance over which the Taylor vortex develops before impact.
Grid and time-step sensitivity check (baseline and a representative gust case)	**[PENDING:]**	Demonstrates that the reported statistics are converged.

Table 1. DNS solver-resolution details required for full reproducibility (§2.2). The simulations are performed with SOD2D without a subgrid-scale model; the solver-resolution metadata required to substantiate the DNS designation are listed here and will be supplied before submission. Values marked pending are owned by the simulation collaborators and are filled from `paper/dns_metadata.yaml` before submission.

that the LEV roll-up and shedding produce. The choice is deliberate: the wake observables can change sharply while the scalar forces change little, so they are the part of the encounter most likely to separate a representation that has captured the gust-vortex physics from one that has only captured its force signature. On these grounds we fix two co-primary endpoints in advance: the wake enstrophy, which the reduced state must carry beyond the release parameters, and the lift coefficient C_L, which the sequential wall-pressure estimator of §3.3 tracks and in which the operating envelope is stated. The remaining three observables are reported as secondary and descriptive. This pre-registration is what the family-wide multiplicity correction in Appendix A is applied against.

The five observables are defined on each frame of the masked vorticity field ω_z described above. The lift and drag coefficients C_L, C_D are as defined earlier in this section. On the wake window $\Omega_w = \{(x, y) : x/c \in [0.5, 4],\ |y/c| \le 1\}$ the wake enstrophy is

$$E_w(t) = \int_{\Omega_w} \omega_z^2 \,\mathrm{d}A, \tag{2.2}$$

and the signed wake circulations integrate the vorticity over its strongly rotational parts,

$$\Gamma^+(t) = \int_{\Omega_w,\ \omega_z > \omega_c} \omega_z \,\mathrm{d}A, \qquad \Gamma^-(t) = \int_{\Omega_w,\ \omega_z < -\omega_c} \omega_z \,\mathrm{d}A, \tag{2.3}$$

with threshold $\omega_c = 1$ in the cleaned vorticity units. The signed circulations at $\omega_c \in \{0.5, 1, 2\}$ are nearly collinear (pairwise Pearson well above 0.9 across held-out frames) and the predictive latent's circulation closure advantage over the reconstructive baseline is essentially unchanged across them, so the closure does not depend on this threshold. Every case uses the same fixed window and mask, and values are reported in the cache vorticity units integrated over chord-normalised area. The forecast horizon $H = 16$ is the sixteenth cached frame after impact, $t = 16\,\Delta t = 0.8\, c/u_\infty$.

One definitional point bounds how these observables are read. The wake and flow observables (the enstrophy and the two circulations) are computed on the same mid-

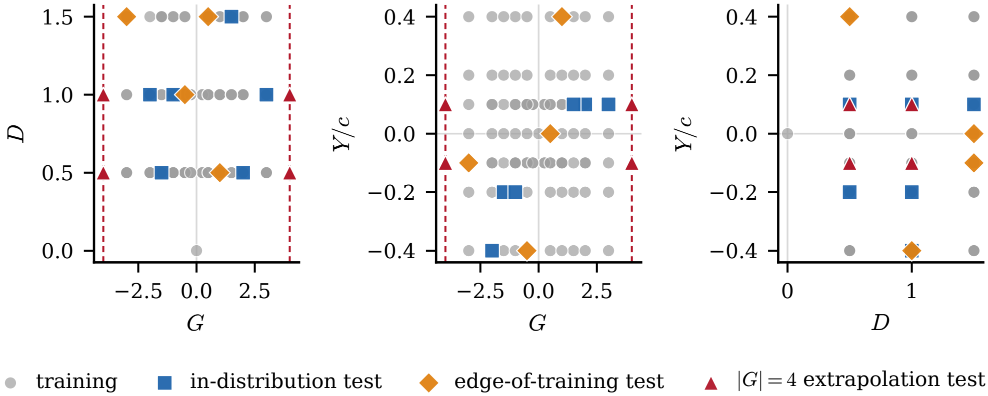


Figure 2. Parameter-space sampling of the 102 cases in the (G, D, Y) gust envelope, shown as three two-dimensional projections and coloured by split: training (268 encounters), the stratified in-distribution test set (42 encounters from 10 cases, split into in-distribution and edge-of-training tiers), and the $|G| = 4$ extrapolation set (40 encounters from 8 cases). The dashed lines mark the $|G| = 4$ extrapolation boundary, now sampled at both signs of the gust so that the boundary set is symmetric rather than one-sided. Training mass concentrates near $Y/c = 0$, which is why the wall-normal offset is the least well resolved parameter (§4.3).

plane slice the encoder consumes, so they omit the out-of-plane content quantified in §2.1 (roughly a fifth of the ω_z enstrophy in distribution), whereas C_L and C_D are true three-dimensional surface-integrated forces. The closure therefore compares mid-plane wake diagnostics against three-dimensional forces, and the mid-plane proxy is weakest at the $|G| = 4$ extrapolation boundary, where the out-of-plane fraction rises substantially.

3. Reduced states, forecasting and wall-pressure estimation

3.1. *Model and training objective*

The model takes no gust parameters: neither the encoder nor the predictor receives (G, D, Y). The predictive encoder E_θ maps the mid-plane vorticity field $\omega_z(192, 96)$ to a latent $\mathbf{z} \in \mathbb{R}^d$ at the headline dimension $d = 32$. It is a hybrid network: three convolutional downsampling stages produce a $24 \times 12 \times 256$ feature map, a six-layer vision transformer processes the resulting tokens, and the class-token output is projected to $\mathbf{z}$ through a single linear layer with a batch-normalisation boundary, which the characteristic-function regulariser below requires. The encoder is therefore a pure state map, and this is the property that lets the latent itself serve as the observable a controller would have access to, since at deployment the gust parameters are not known a priori (§5).

The predictor P_ϕ advances the pooled state in time, from its own history alone, so the model is unconditioned end to end. It is a vector sequence model: a six-layer autoregressive transformer with rotary temporal positions and a causal mask that maps a short history of 32-vectors to the next 32-vector, with no gust input. From the two most recent encoded states it rolls eight steps open loop, feeding its own predictions back in, and the loss is the multi-step error between the rolled coefficients and the online encoder outputs at the matching frames, detached. The pipeline is therefore as plain as a linear or autoencoder reduced-order model: 32 numbers in, a dynamics model on the vector, 32 numbers out, with no spatial map anywhere in the loop. The state is the pooled vector at every stage: the anti-collapse regulariser, the observable heads, the probes, and the estimator all read the same 32-vector, and nothing in the model carries more. The estimator of §3.3 refits a predictor of this same class on the frozen pooled trajectories for its forecast step, so the training and estimation dynamics share a function class. Internal structure for both is given in Appendix A. Figure 3(a) shows the encoder and predictor, and panel (b) sets out the matched-predictor evaluation protocol used throughout; the contrast between the predictive training signal and the reconstructive one, in which the reconstructive autoencoder passes the field at time t through an encoder and a decoder and incurs a field-space error whereas the predictive model encodes the fields at t and $t + \Delta t$ with shared weights and incurs a latent-space error between the predicted and observed target embeddings, with no field-space loss, is drawn in Appendix A (figure 14). The two networks are an unconditional hybrid convolution-ViT encoder mapping the ω_z field to the pooled $d = 32$ state and a six-layer causal transformer predictor rolling that state; their full configuration, widths and parameter counts are in table 11 in appendix A.3.

The predictive encoder and predictor are trained jointly on a multi-step open-loop rollout term, the predictor rolled eight steps from its own outputs with no teacher forcing against detached online targets, and an anti-collapse regulariser that prevents the encoder from collapsing to a constant. We use the characteristic-function regulariser of Balestriero & LeCun (2025); Maes *et al.* (2026), which matches the latent distribution to an isotropic Gaussian through a small number of random projections, with an automatic fall-back to a variance-covariance objective (Bardes *et al.* 2022) should the participation ratio of the latent indicate collapse. Its configuration is given in Appendix A.

Three auxiliary heads read aerodynamic observables off the encoder output during training: a lift head that maps $\mathbf{z}_t$ to $C_L(t)$, a wake head that maps $\mathbf{z}_t$ to an 80-dimensional signed wake spectrum, and a near-body head, constructed in §3.1.1 below, that maps $\mathbf{z}_t$ to an 80-dimensional lift-element observable of byte-matched capacity. All are trained with small weights jointly with the representation and none is part of the latent-prediction loss. The scalar-lift lineage is the observable augmentation of Fukami & Taira (2023, 2025); which head supplies which property of the state is the subject of the conditioning study of §4.1.1. The wake target itself is a fixed descriptor chosen by design rather than tuned: it pairs a coarse signed spatial pooling of the wake region, which retains the layout of the alternating positive and negative vortices while discarding pixel-level detail, the spatial-pyramid construction of Lazebnik *et al.* (2006), with a radially-averaged power spectrum, the canonical isotropic scale descriptor of a turbulent field (Pope 2001) and an established compact image descriptor in machine learning (Durall *et al.* 2020), so it encodes both where the wake vortices sit and at which scales their energy resides. Pairing a localised spatial summary with a multiscale spectral one is the spatial-plus-scale representation formalised by the wavelet scattering transform (Bruna & Mallat 2013), which has been applied to turbulent flow fields precisely to recover geometry that the power spectrum alone cannot resolve (Skinner *et al.* 2025). We keep the descriptor compact, at 80 dimensions, so that it constrains the wake structure without over-parameterising the target on the available training encounters. The head-supervision axis is no longer a confound: the $2 \times 2 \times 2$ conditioning cube of §4.1.1, three seeds per cell with matched reconstructive anchors, isolates each head's contribution under gates fixed in advance of evaluation.

Comparing a lift observable against a wake observable is only meaningful if the comparison is fair, and five choices make it so. First, both field-observable targets are 80-dimensional and byte-matched, built by the identical construction, so no head sees a richer training signal. Second, the heads share one network class and one loss weight, so the comparison isolates the supervision signal and not the head architecture. Third, the reconstructive baselines carry the identical heads, the AE-L, AE-W and AE-LW anchors of §4.1, so supervision is never an advantage of the predictive model alone and every supervised-versus-unsupervised and predictive-versus-reconstructive contrast is available separately. Fourth, the wake descriptor is assembled from established, fixed constructions, the spatial pyramid and the radial power spectrum above, and was not tuned against any outcome. Fifth, the conditioning gates were written and archived before any result was read, three seeds per cell, with probes fit on the training split and scored on the held-out split alone. The one asymmetry a reader might raise, that the linear basis carries no supervision channel, is by design: supervision is a property of trainable encoders, so the proper orthogonal basis is the objective-free linear reference, read through the identical frozen probes, and the matched-supervision autoencoder exists precisely so that objective and supervision are separable.

3.1.1. *The near-body lift-element head*

The third supervision head targets a physically constructed near-body observable built on the force-element decomposition of Chang (1992), in the form popularised for vortex-dominated flows by Menon & Mittal (2021). An auxiliary potential ϕ_L is defined by the exterior boundary-value problem

$$\nabla^2\phi_L = 0 \ \text{ outside the body}, \qquad -\mathbf{n}\cdot\nabla\phi_L = \mathbf{n}\cdot\mathbf{e}_L \ \text{ on the body}, \qquad \phi_L \to 0 \ \text{ far away}, \tag{3.1}$$

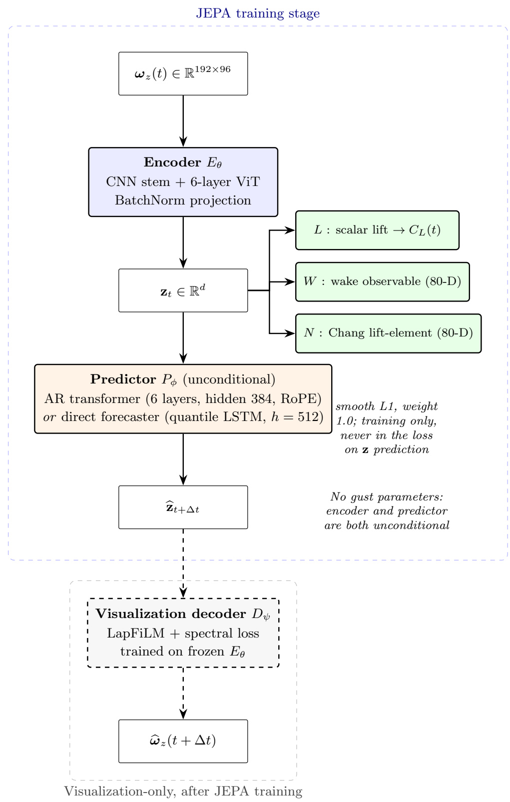
with $\mathbf{e}_L = (-\sin\alpha, \cos\alpha)$ the lift direction at $\alpha = 14°$. Equation (3.1) is discretised once on the 192×96 cache grid with a staircase immersed Neumann condition on the solid-adjacent cell faces and a homogeneous Dirichlet far field, and solved directly (figure 21); the interior Laplacian residual of the stored solution is 6.4×10^{-13} in the maximum norm. The lift-element density is then the pointwise field

$$e(\mathbf{x}, t) = \omega_z\left(-v\,\partial_x\phi_L + u\,\partial_y\phi_L\right), \tag{3.2}$$

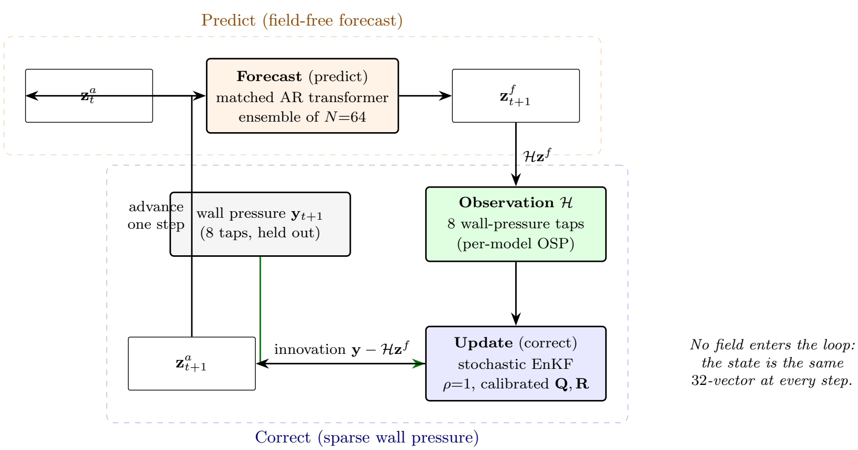
evaluated from the raw mid-plane velocity and vorticity with the stored sign convention verified empirically (the cache stores $\partial u/\partial y - \partial v/\partial x$, so the standard ω_z is its negation). Because the cache window truncates the domain, the full-domain integral of (3.2) does not track the lift; the band-restricted field does, and the head therefore supervises e inside a feathered near-body band $w(\mathbf{x}) = \max\{0,\ 1-\delta(\mathbf{x})/0.3c\}$, with δ the Euclidean distance to the body and its adjacent cells. The banded field $w\,e$, scaled by a fixed constant chosen so its 99th percentile magnitude is order one, is compressed to an 80-dimensional target by exactly the wake-observable construction: 64 sign-preserving patch energies (an 8×4 grid of $\log(1+\mathrm{relu}(\pm x)^2)$ pooled energies) concatenated with a 16-bin Hann-windowed radial power spectrum. The two 80-dimensional targets are byte-matched in capacity, and the wake and near-body heads share one network class, so the conditioning comparison of §4.1.1 isolates the supervision signal, not the head architecture. A precompute quality gate requires the median lagged correlation between the band-integrated element and the stored lift over the gust-bearing training encounters (lag search $|\ell| \le 25$ frames) to exceed 0.4; the achieved value is 0.74. The principled weighting is not interchangeable with a plain vorticity-magnitude proxy: the patch-block cosine between the two targets is 0.68 on the comparison sample. The auxiliary potential and its supervision band are shown in figure 21 in appendix A.2.

3.2. *Matched-predictor closure protocol*

The comparison rests on a single principle: every baseline is the same pipeline, and only the encoder differs. Each encoder produces a latent trajectory $\mathbf{z}_{1:T}$ from the vorticity fields, and an identical probe family maps a latent to a physical observable for evaluation (figure 3(b)). Every latent is a pooled coefficient vector at the same dimension, so the same two forecast operators apply to all of them: a matched autoregressive transformer that reads the vector sequence directly, and a matched residual U-Net that tiles the vector to a grid



(a)



(b)

Figure 3. The framework, from representation to deployment. (a) The encoder maps the mid-plane vorticity field to a latent $\mathbf{z}_t$, and the predictor, in either of its two variants (the vector transformer of table 11 or the direct multi-horizon forecaster of §3.2.1), advances the latent from its own history; no gust parameters enter the encoder or the predictor, the three supervision heads (L, W, N) read $\mathbf{z}_t$ during training with small weights and are never part of the latent-prediction loss, and the visualisation decoder is trained only after the encoder is frozen. (b) At deployment the pressure-only predict-correct estimator advances an ensemble of 64 members field free, then corrects them from 8 span-averaged wall-pressure taps; the innovation is the measured minus the predicted observation, in tap space for the base filter of §3.3.1 and in encoded latent space for the filters of §3.3.2 and §3.3.3. No vorticity field enters the loop and the lift and wake probes never enter the innovation. The evaluation protocol, the matched forecast operator and probe family applied identically to every encoder, is figure 20 in appendix A.

and forecasts it convolutionally, both fitted with the same recipe and budget. They are not interchangeable in practice: the transformer is numerically stable on the pooled states of the controlled matrix but diverges when refitted on the reference latents, collapsing to a negative merit (Appendix A), while the U-Net is stable on all. We therefore report each latent under the operator that is stable on it, the transformer on the pooled states and the U-Net on the reference latents, rather than force one operator and either penalise the pooled states off-class or break on the references. The wake-supervised predictive state's own co-trained predictor, the co-trained reduced-order model, is reported alongside, and the sequential estimator of §3.3 uses the same matched transformer on the wake-supervised predictive state's frozen trajectories. Every state is compared at the matched headline dimension $d = 32$; the dimension plateau at $\{16, 32, 64\}$ and the minimum-dimension panel of §4.3 establish that this choice is a robustness statement rather than a tuned operating point. This isolation is the only fair way to attribute representation quality to the encoder rather than to incidental differences in the readout. The reconstructive references use their best-documented configuration, with ReLU activations and group normalisation rather than the strict published variant (tanh, no normalisation), which we found gives a worse parametric probe; the comparison is therefore against a tuned baseline, not a hobbled one. The observable heads are pinned to the same current-frame form for every family, so no family gains an accidental advantage from a differently-shaped supervision signal. The transformer-or-U-Net split above is retained as the legacy suited-operator protocol; the primary forecast-merit comparison of §4.4 uses one further operator, trained identically on every family's latents, so the family ordering cannot be an artefact of operator choice.

3.2.1. *The shared direct multi-horizon forecaster*

The forecast-merit comparison of §4.4 requires one operator trained identically on every family's latents; the operator used throughout is the direct multi-horizon forecaster, a quantile forecaster whose design transfers the deployable ingredients of the TiRex forecasting line (Auer *et al.* 2025; Podest *et al.* 2026) without its foundation-model machinery. Each context window of L consecutive states (L drawn uniformly from 16 to 30 during training, fixed at $L = 25$ at evaluation) is normalised per window, $\tilde{\mathbf{z}} = \operatorname{arcsinh}((\mathbf{z} - \mu)/\sigma)$ with μ, σ the window mean and standard deviation (floored at 10^{-3}), passed through a two-layer LSTM of width 512, and mapped by a single feed-forward head to the full forecast tensor of 40 horizon steps, d coordinates and nine quantiles $q \in \{0.1, \ldots, 0.9\}$ in one shot; the inverse transform restores physical scale. There is no autoregressive feedback, so rollout error cannot compound by construction. Training minimises the pinball loss

$$\mathcal{L} = \frac{1}{|Q|} \sum_{q \in Q} \left\langle \rho_q\left(z - \hat{z}_q\right) \right\rangle, \qquad \rho_q(u) = \max(qu,\ (q-1)u)\,, \tag{3.3}$$

with AdamW. The median forecast is the point prediction everywhere; the predicted inter-quantile band supplies the state-dependent process noise of §3.3.3. The optimiser settings, the one-off backbone and width selection (LSTM against an sLSTM cell (Beck *et al.* 2024) at two widths and two quantile counts, the sLSTM tying within seed noise), and the adopted and rejected ingredients of the TiRex source line, including the oracle-covariate negative of §4.4, are catalogued in the supplementary material.

The headline figure of merit is representational closure: a fixed linear probe is applied to the latent encoded directly from a held-out field, and we ask whether it recovers each of the five observables, the wake in particular. This isolates what the encoder carries, with no rollout. We report it as the held-out coefficient of determination R^2 on the in-distribution test and boundary test sets, with the per-observable mean absolute error alongside, and bootstrap 95% confidence intervals over encounters. As a secondary, confirmatory measure we also report forecast closure: the predictor is rolled recursively from the pre-impact context to $H = 16$, and the same probe maps the predicted latent to each observable. A small forecast error means the encoder produces a latent whose dynamics, propagated by the predictor, retain the information the probe needs; a large error means either that the encoder discarded that information or that the predictor lost it during rollout, a distinction the drift diagnostic below resolves. The representational closure is in table 2, which reports the readability and the matched-predictor forecast merit side by side. We do not headline a forecast-R^2-versus-horizon curve; the rollout is reported to show that the latent stays usable, not as a validated forecast.

3.3. *Estimating the state from the wall*

At deployment the vorticity field is unavailable and the wall pressure is not, by itself, enough to fix the state: a single pressure frame recovers little of the wake through any of the latents (§4.5). The reason a single frame is not enough, and a window is, is a statement about the dynamics rather than about the number of sensors. The encounter lies on an attractor of low effective dimension, a limit cycle with a gust excursion and a relaxation, into which the encoded latent concentrates its variance (§4.3). Delay-embedding theory (Takens 1981; Sauer *et al.* 1991) motivates reading the wall pressure over a window of delays rather than at a single instant, since a generic observation of a low-dimensional system reconstructs its state up to a diffeomorphism once the product of tap count K and delay count m exceeds twice the attractor dimension. Its generic guarantees do not directly establish sufficiency for the finite, parametrically forced transient family considered here, so we take the theorem as an organising principle and treat the sensor-delay map of §4.5 as the empirical evidence; there, a handful of taps over a modest window suffices and the delay window substitutes for sensor count. A coordinate the instantaneous pressure misses, the wake circulation, is still reconstructed by the window provided it is dynamically coupled to what the pressure sees, which for the coupled vortex dynamics it is. The estimator must therefore be sequential, accumulating pressure innovations through learned dynamics, which is the setting of data assimilation.

Every estimator in this paper solves the same discrete state-estimation problem on the frozen coefficient state. Writing $\mathbf{z}_t \in \mathbb{R}^d$ for the encoded state at frame t (time base $\Delta t = 0.05\, c/u_\infty$, the cache frame spacing) and $\mathbf{p}_t \in \mathbb{R}^K$ for the wall pressure at the K selected taps, the pair

$$\mathbf{z}^f_{t+1} = F\left(\mathbf{z}^a_t\right) + \boldsymbol{\eta}_t, \qquad \boldsymbol{\eta}_t \sim \mathcal{N}(0, Q_t), \tag{3.4}$$

$$\mathbf{y}_t = h(\mathbf{z}_t) + \boldsymbol{\epsilon}_t, \qquad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, R), \tag{3.5}$$

defines each member of the suite once its four ingredients (F, h, Q_t, R) are named; the remainder of this subsection does exactly that, and table 15 collects every parameter. The frozen probes that read the lift and the wake enstrophy off the state can never enter the innovation: this leakage guarantee is enforced by construction and by a unit test, so the reported analysis closure is a genuine estimate, not a quantity the filter was given (figure 3bb). Two evaluation frames are used and always stated. The envelope frame assimilates over $[t_{\mathrm{imp}} - 24,\ t_{\mathrm{imp}} + 48]$ frames with a field-free initialisation at the window start; the ladder frame runs the full encounter with the ensemble initialised from the encoded observation at $t_0 = 9$ (the delay-embedding warm-up) and scores on the same impact window. All estimator fits, without exception, use the training split only.

3.3.1. *Base stochastic ensemble filter*

The frozen production filter is a stochastic ensemble Kalman filter (Evensen 2003) with $N = 64$ members. The forecast model F is the matched autoregressive transformer of §3.1, refitted on the frozen pooled training trajectories; each member carries its own buffer of past analysis states (at most 32 frames) and is advanced one step per frame. The process noise is state-independent, $Q_t \equiv Q$, estimated as the sample covariance of one-step teacher-forced predictor residuals over 256 training sub-trajectories of length 32, with a 10^{-6} diagonal floor. The observation map h is a ridge-regularised linear read-out from the state to the K taps (penalty 1.0), fitted on four fifths of the training encounters; the observation-noise covariance R is the diagonal of the residual variance of that map on the held-out fifth, grouped by encounter, with floor 10^{-6}. With ensemble mean $\bar{\mathbf{z}}_t^f = N^{-1}\sum_i \mathbf{z}_t^{f,(i)}$ and deviations $X_t^f = [\mathbf{z}_t^{f,(i)} - \bar{\mathbf{z}}_t^f]_i$, the update is the perturbed-observation analysis

$$P_t^f = \frac{X_t^f (X_t^f)^\top}{N-1}, \qquad K_t = P_t^f H^\top \left(H P_t^f H^\top + R\right)^{-1}, \tag{3.6}$$

$$\mathbf{z}_t^{a,(i)} = \mathbf{z}_t^{f,(i)} + K_t \left(\mathbf{y}_t + \boldsymbol{\epsilon}_t^{(i)} - H\mathbf{z}_t^{f,(i)}\right), \qquad \boldsymbol{\epsilon}_t^{(i)} \sim \mathcal{N}(0, R), \tag{3.7}$$

where H is the Jacobian of the linear h. Multiplicative inflation $\mathbf{z}^{f,(i)} \leftarrow \bar{\mathbf{z}}^f + \rho\,(\mathbf{z}^{f,(i)} - \bar{\mathbf{z}}^f)$ is applied before the update; ρ is selected per family on the validation split only ($\rho = 1.00$ for the predictive and linear families, $\rho = 1.05$ for the reconstructive one). The ensemble is initialised at the window start from a windowed pressure-to-state regressor (a Nyström radial-basis ridge with 300 components on a 12-frame causal pressure window) by sampling its training-residual covariance around the regressed mean, so no field information enters the filter at any time. The frozen observable probes that score the filter never enter the innovation: the only measured quantity in (3.7) is the K-tap wall pressure.

3.3.2. *The linear latent filter with encoded observations*

The structural-consistency retrofit, adapted from the latent-space assimilation construction of Tong *et al.* (2026), replaces both F and h. The dynamics are linear, $F(\mathbf{z}) = A\mathbf{z}$, with A the ridge solution of $\mathbf{z}_{t+1} \approx A\mathbf{z}_t$ over all training transitions (penalty 1.0) followed by a projection of its singular values onto $[0, 1]$, the hard analogue of their spectral hinge penalty; Q is the covariance of the resulting fit residuals. The observation side replaces the sparse taps by an encoded full-state observation: a delay-embedding operator lifts the normalised tap history $\mathbf{p}_{t-m+1:t} \in \mathbb{R}^{Km}$ with $m = 10$ (causal, edge-padded), and a ridge map W (penalty 1.0) regresses it onto the state, so that

$$\mathbf{y}_t = E_{\mathrm{obs}}(\mathbf{p}_{t-m+1:t}) = W^\top \operatorname{vec}(\mathbf{p}_{t-m+1:t}), \qquad h = \mathrm{Id}, \qquad R = \mathit{\Gamma}, \tag{3.8}$$

with $\mathit{\Gamma}$ the covariance of the E_{obs} training residuals plus a 10^{-8} jitter. The same E_{obs}, evaluated framewise with no dynamics at all, is the static delay-embedded inverse that anchors the estimator ladder. Filter consistency failure is an ensemble-consistency diagnostic, declared by one fixed criterion everywhere in the paper: the normalised innovation squared exceeds the $\chi^2_{0.99}$ quantile at the observation dimension for five or more consecutive frames, or any analysis Mahalanobis ratio exceeds 3. It flags an under-dispersed or inconsistent ensemble and is distinct from a large-error encounter, for which the analysis lift itself fails (impact-phase R^2 below -1); a filter can post consistency failures while its mean still tracks the load. The observation-rate protocol subsamples the update, not the sensor: pressure is recorded every frame and the analysis step is applied every m-th frame only.

3.3.3. *Forecast-derived state-dependent process noise*

The forecast-noise filter keeps the encoded observation (3.8) and replaces the member propagation by the direct multi-horizon forecaster of §3.2.1: each member is advanced by the one-step median forecast, and the process noise is read off the forecaster's own uncertainty band at forecast time,

$$\mathbf{z}_{t+1}^{f,(i)} = \hat{\mathbf{z}}_{t+1}^{q_{0.5}} + c\,\boldsymbol{\sigma}_{t+1} \odot \boldsymbol{\xi}^{(i)}, \qquad \boldsymbol{\sigma}_{t+1} = \frac{\hat{\mathbf{z}}_{t+1}^{q_{0.9}} - \hat{\mathbf{z}}_{t+1}^{q_{0.1}}}{2 \times 1.28155}, \qquad \boldsymbol{\xi}^{(i)} \sim \mathcal{N}(0, \mathrm{Id}), \tag{3.9}$$

so $Q_t = c^2\,\mathrm{diag}(\boldsymbol{\sigma}_{t+1})^2$ is state-dependent through the predicted inter-quantile width (the denominator converts the $q_{0.9} - q_{0.1}$ band of a Gaussian to its standard deviation). The single scalar c is the filter's only tuned quantity and it carries two independent, validation-only calibrations. The production value $c = 1.77$ makes the one-step $q_{0.1}$–$q_{0.9}$ band cover 80 % of held-out training transitions. A second calibration, fixed in advance, before any value was seen, selects c on the held-out encounters of the training cases by matching the pooled impact-phase normalised innovation squared, $\mathrm{NIS}_t = \nu_t^{\top} S_t^{-1} \nu_t / d$ with $\nu_t = \mathbf{y}_t - \bar{\mathbf{z}}_t^f$ and $S_t = P_t^f + \Gamma$, toward its consistency expectation of one, over the fixed grid $c \in \{1.0, 1.4, 1.77, 2.5, 3.5, 4.5, 6.0\}$, followed by a single frozen run on the test splits. The outcome of that calibration, reported in §4.6.3, is itself a result. The two-stage filter applies the encoded-observation update (3.8) with the median-forecast propagation (3.9) inside the impact window only, and outside it propagates members with the matched transformer and assimilates in tap space through the linear map of §3.3.1; the schedule boundary is the frozen impact-window mask of the cache and is not tuned.

3.3.4. *Fixed-lag smoother*

On the linear stack of §3.3.2 a fixed-lag Rauch–Tung–Striebel smoother provides the reanalysis reference. With forecast and analysis moments $(\mathbf{z}_k^f, P_k^f)$ and $(\mathbf{z}_k^a, P_k^a)$ from the forward Kalman pass, the backward recursion over the lag window is

$$C_k = P_k^a A^{\top} \left(P_{k+1}^f\right)^{-1}, \qquad \mathbf{z}_k^s = \mathbf{z}_k^a + C_k \left(\mathbf{z}_{k+1}^s - \mathbf{z}_{k+1}^f\right), \tag{3.10}$$

applied over a lag of $L = 5$ frames, that is a fixed reporting latency of $0.25\,c/u_\infty$; the smoothed estimate at t uses pressure up to $t + L$ and nothing later. The ensemble smoother variant on the forecast-noise filter (a lagged cross-covariance update of the stored member history) degrades the filter and is reported as a negative in the supplementary material.

3.3.5. *Sensor placement*

Each family senses at its own taps. The per-family placement is a greedy forward selection over the 192 physical wall taps: at each step the candidate tap whose 30-frame causal pressure window, concatenated to the already-selected windows, best ridge-predicts the leading principal component of the family's own impact-frame states is added. Writing r_j for the ridge residual mean square of the joint feature set extended by tap j, and $g_j = \max\{0,\, r_\varnothing - r_j\}$ for its reduction over the target variance, the selected tap maximises

$$J(j) = n\,g_j - \tfrac{1}{2}\,g_j - \tfrac{1}{2}\,n\,r_j + \tfrac{1}{2}\,\frac{n\,g_j}{g_j + 10^{-6}}, \tag{3.11}$$

with n the number of training encounters; at this configuration the first and third terms dominate, so the criterion is explained-variance gain penalised by the residual. The staircase is nested by construction, each budget warm-starts from the previous selection, so the $K = 16$ array contains the $K = 8$ array. The shared target-blind reference array is the column-pivoted QR (qDEIM) selection of Drmač & Gugercin (2016) on the leading pressure modes. Which placement a given figure uses is stated in its caption. The per-estimator configuration is collected in table 15 in appendix A.3.

The mixed timescales of this estimator, a transition map trained on sequences and an emission map trained per instant, mirror the structure of the training objective in §3.1: a dynamics equation is a statement about trajectories and an observation equation is a statement about single instants. The combined loss is a jointly trained state-space realisation, and the filter is the same decomposition read at inference.

3.4. *Diagnostics, controls, and uncertainty*

Two diagnostics turn the closure result into a mechanism. The first is the latent drift, which addresses the fact that a predictor with low one-step error need not produce a useful long rollout, because each prediction is fed back as the next input and small geometric errors accumulate. We quantify how far a rollout has left the training manifold by the Mahalanobis distance of the rolled-out latent at horizon H, measured against the distribution of simulation-encoded latents for the same encoder, normalised by the Mahalanobis distance of the simulation-encoded latent itself. A ratio near one means the rollout stays within the natural spread of the training latents; a large ratio means it has gone out of distribution, so that the probe is then queried outside its support.

The second diagnostic is a parameter-only floor, which matters here because the model never sees the gust parameters: it bounds what the parameters alone could explain, so that any closure above it is genuinely supplied by the latent state. We compute it with a kernel-ridge regressor that maps $\mathbf{c} = (G, D, Y)$ directly to each observable at $H = 16$, with no latent input, fit on training and evaluated on the held-out splits. On wake enstrophy this floor is low, so the latent's wake closure is not something the parameters could have supplied. At the impact frame the same parameter floor is high, because the just-released gust nearly determines the instantaneous state; the gap opens as the wake evolves (§4.1.2).

Because the predictive encoder differs from the reconstructive baseline in objective, architecture, and auxiliary supervision at once, the comparison is accompanied by controls that vary one factor at a time (table 12): a crossing of objective with architecture at matched auxiliary heads, an ablation of each auxiliary head, and a conditioned predictor variant that adds $\mathbf{c}$ (§4.4.1). All the controls share the one encoder of table 11 and differ only in what is trained on top of it. The objective-free supervised control isolates the objective most directly: it trains the shared encoder with the wake and lift heads and the anti-collapse term but with no predictive or reconstructive objective, no predictor and no decoder, so it holds the supervision fixed and removes every objective, isolating what the observable heads alone supply. The published reference recipes, a plain convolutional autoencoder in the Fukami–Taira lineage (Fukami & Taira 2023, 2025), the β-variational autoencoder of Solera-Rico *et al.* (2024), and a linear proper-orthogonal-decomposition basis, use their own native architectures but produce a pooled coefficient vector at the same latent dimension. The forecast merit refits a matched operator on each frozen latent; we use the operator that is numerically stable on it, a transformer on the pooled states and a residual U-Net on the reference latents, on which the transformer diverges (§3.5, Appendix A).

Every held-out R^2 and mean absolute error reported below carries three independent uncertainty signals, following the reporting protocol fixed with the data split: a 2000-resample bootstrap over the held-out encounters, the standard deviation across three encoder retrains from independent seeds, and a five-fold cross-validation over cases on the probe read-out. Their full specification is in Appendix A.

3.5. *Protocol, phases and uncertainty accounting*

Every encounter is scored in three phases anchored to its impact frame: lead-in $[t_{\mathrm{imp}} - 8, t_{\mathrm{imp}})$, impact $[t_{\mathrm{imp}}, t_{\mathrm{imp}} + 16)$ and relaxation $[t_{\mathrm{imp}} + 16, t_{\mathrm{imp}} + 48)$, all half-open in frames and clamped to the encounter. Load metrics are reported in three complementary forms because each corrects a failure mode of the others: the coefficient of determination $R^2 = 1 - \mathrm{SSE}/\mathrm{SST}$ pooled over the phase frames, the root-mean-square and mean absolute errors in lift-coefficient units, and peak-relative quantities. The peak diagnostics locate the extremum of $|C_L|$ within the impact phase in the estimate and in the truth separately and report the value difference, its magnitude as a percentage of the true peak, and the timing difference in convective units at $\Delta t = 0.05$. Peak-region readability uses a pooled R^2 over windows of half-width 8 frames centred on each encounter's true peak, with a persistence floor that holds the mean of frames $[0, 25)$ constant; phase lag is the argmax of the normalised cross-correlation within ± 20 frames with parabolic sub-frame refinement, positive when the estimate trails the truth. Decoded fields are scored by the structural similarity index in the convention of Wang *et al.* (2004), $K_1 = 0.01$, $K_2 = 0.03$, with the data range pinned per dataset version to twice the global 99.9th percentile of the normalised target magnitude ($L = 8.487$ for the present split), on three masks: the feathered near-body band of §3.1.1, a wake window spanning $0 < x/c < 4.5$ and $|y/c| < 1.25$, and the full frame. Linear-probe R^2 is used throughout as a readability metric and never as an information metric; the control that motivates this distinction, a multilayer probe that equalises what a linear read-out cannot reach, is reported alongside every dimension comparison (§4.2.2). When an observable is read at a specific instant of the encounter rather than over the window, the same linear probe, fit on the training window, is evaluated on the held-out encounters at the pre-impact, impact and peak-lift frames (± 2), and reported both as its R^2, with the mean observable at that instant as the baseline, and as its physical error, the root-mean-square deviation in the observable's own units; the two are complementary, the error exposing a large absolute miss where the high cross-encounter variance at a critical instant flatters the R^2.

Uncertainty is accounted at three levels and every stochastic number in the paper carries its count. Encoder training variance uses three seeds wherever a cell is compared (the conditioning cube, the $d = 4$ bands, the direct-predictor and reconstruction-baseline bands); single-seed cells appear only in the dimension grid and are disclosed as such in each caption. Filter member-noise variance uses five seeds of the perturbed observations and initial ensemble. Paired family comparisons use a case-clustered bootstrap that resamples whole cases with replacement (2000 resamples on encounter means, 10^4 on case means) and reports the interval that excludes or straddles zero; multiple comparisons within one question family are Holm-corrected. The acceptance gates for the conditioning study and for the present campaign's follow-up runs were fixed in advance of evaluation: written and committed before any corresponding result was read, with both declarations shipping with the data record. The gust-intensity axis is reported in the inventory sign convention, and no test-C quantity is used for any selection anywhere in the pipeline.

4. Results

We evaluate three families of reduced-order state at matched pooled dimension: a predictive coefficient state, a matched-supervision reconstruction autoencoder, and a linear proper orthogonal basis, with the published-recipe reconstruction and a β-variational autoencoder retained as reference recipes. The controlled representational comparisons (§4.1 to §4.5) are reported on the in-distribution test encounters, with the boundary test set held in reserve so that the $|G| = 4$ extrapolation enters only as a final reported number and never as a selection signal. The estimator and the operating envelope (§4.6) are characterised with a frozen filter across all encounters, training, validation, in-distribution test and boundary test sets alike, and the per-encounter paired tests on the in-distribution test and boundary test sets are given in Appendix A. Every quantity quoted below is a held-out value at the readout horizon unless stated otherwise. One convention holds throughout: the wake-supervised predictive state (conditioning-cube cell CLW) is the single predictive representative carried into the forecasting and estimation comparisons, and the lift-focused predictive state is the lift-sharpened variant whose accuracy ceiling the construction section quantifies.

4.1. *A non-collapsed predictive state*

4.1.1. *Constructing the state: which supervision supplies which property*

The construction question is answered by a $2 \times 2 \times 2$ cube over the three supervision heads $\{L, W, N\}$, trained at three seeds per cell on the pooled pipeline of §3.1 with matched reconstructive anchors, under gates written and archived before any result was read; probes are fit on train and scored on the held-out in-distribution test encounters, the validation set steering nothing and the boundary test set staying in reserve per the policy above. Under the predictive objective the scalar lift head is necessary for latent health. Every cell without L collapses, with participation ratios of 2.50 (C0) and 2.55 (CW) against a floor of 9.6, flat from the first diagnostic; neither the wake observable (figure 4) nor the near-body observable (appendix A.1), nor both together, substitutes for the scalar anchor. The reconstruction objective anchors without it: the wake-only autoencoder keeps a healthy latent (participation ratio 21.94) yet reads the peak loads poorly ($R^2 = 0.47$), so a healthy latent and a lift-readable latent are different achievements. The collapse is the low-intrinsic-dimension failure mode anticipated for the characteristic-function regulariser at its default configuration; the automatic variance-covariance fallback is armed at an iteration threshold the pooled runs never reach, so the cube exposes the mechanism rather than masking it.

Nothing replaces the lift anchor. Replacing L by W fails decisively, the wake-only cell trailing the lift-only cell by -35.22 peak-region R^2 points (case-clustered interval [-67.97, -11.42]), and replacing L by N fails the same way (-21.06 [-30.91, -12.33]). On top of L, a near-body force-element head sharpens the peak load further and is the strongest cell in the cube (appendix A.1). We carry the wake-headed wake-supervised predictive state (cube code CLW) as the single predictive representative downstream: its marginal value on top of L is geometry rather than lift, matching the lift-only cell on the peaks (peak-region $R^2 = 0.84 \pm 0.015$ over three seeds against 0.83 ± 0.021 for the lift-only cell, the paired difference straddling zero at [-3.78, 2.75]) while adding near-body structural fidelity (0.017 [0.009, 0.026] SSIM in the Wang convention of §3.5).

The three heads therefore divide the labour: L anchors the latent, N sharpens the load, and W carries the wake state. The wake-enstrophy probe makes the last division quantitative: the wake-supervised predictive state reads the wake enstrophy E_w at $R^2 = 0.73$ linear and 0.84 through a multilayer probe (three seeds, in-distribution test set; this enstrophy-specific probe is distinct from the pooled wake-descriptor readability 0.751 quoted in table 2), where the lift-focused predictive state reaches only 0.37 and 0.63. The anchor comparison is required too: the reconstructive AE-LW carries the most linearly readable wake state of all (0.77 linear, 0.90 MLP), as a reconstruction objective should; the predictive family's wake advantage is not probe readability but what survives forecasting and assimilation, quantified in §4.4 and §4.6.3. The full task-dependent readability matrix, five observables across every cell and anchor, is figure 17 in appendix A: the wake block lights up exactly where wake supervision or a reconstruction objective sits, and nowhere else. Because lift is included during training, lift accuracy alone cannot establish the role of the lift head.

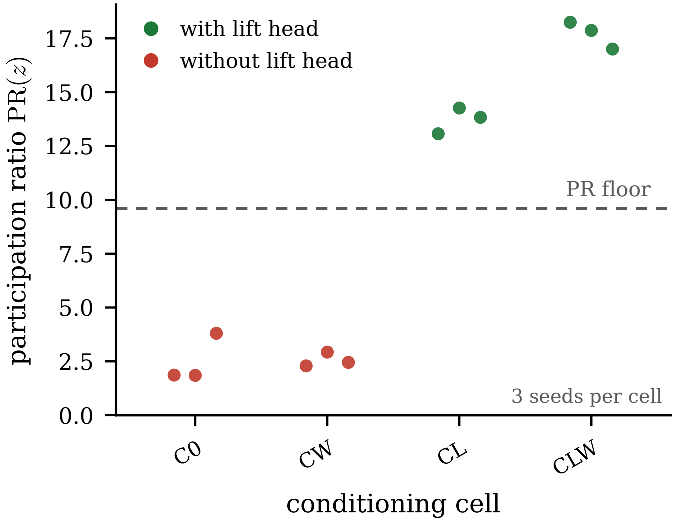


Figure 4. The conditioning cube on the lift and wake axes: final participation ratio per cell (in-distribution test set, three encoder seeds per cell) against the collapse floor $0.3\,d = 9.6$, every no-L predictive cell (C0, CW) collapsing and every L cell (CL, CLW) healthy. The training history of the collapse, flat from the start of training while a healthy cell crosses the floor within the first quarter of the run, is figure 19 in appendix A; the near-body (N) cells and the paired peak-load increments are in appendix A.1.

Its structural role is instead identified from the collapse of every predictive state trained without that head; the incremental effect of the near-body target is evaluated relative to an already lift-supervised state, under paired gates fixed in advance of evaluation. The two claims rest on different comparisons and fail independently.

4.1.2. *What the coefficient state carries, and what supplies it*

The comparison between reduced states is meaningful only if the wake is a discriminating endpoint, and it is. At the readout horizon the release parameters no longer fix the wake enstrophy, while they still fix the forces, so the wake is where a representation that has captured the gust-vortex physics parts company from one that has captured only its force signature. Table 2 reports what each pooled coefficient state carries, read by a fixed linear probe from a held-out in-distribution test field with no rollout. The wake is read cleanly by every state that carries the wake-supervision head, with a held-out R^2 between 0.750 and 0.792, and it collapses for every state that does not, falling to 0.160 for the predictive latent trained without the head. Instantaneous readability of the wake is therefore supplied by the observable supervision, not by any training objective. The lift, read by the same windowed probe, is a different story: every anchored state carries it well, and the predictive state reads the lift at least as accurately as the wake (0.813 against 0.751), so the wake supervision sharpens the wake without costing the load.

The objective-free supervised encoder makes this concrete. Trained with the wake head but no predictive or reconstructive objective at all, it reads the wake at least as well as the full predictive latent (0.792 against 0.751): the paired readability gap is 0.041 in the wake with a confidence interval $[-0.047, 0.162]$ that includes zero, so the instantaneous readability is supplied by the supervision, not by the training objective. Under one shared forecast operator, the direct multi-horizon forecaster of §3.2.1 trained identically on every state, the three wake-headed states are also the most forecastable and are statistically indistinguishable from one another (0.561, 0.574 and 0.443 at horizon sixteen, case-

Table 2. What each pooled coefficient state carries, at matched dimension $d = 32$ on the 42 in-distribution test encounters. Held-out wake-enstrophy readability and, on the same windowed linear probe, lift readability (both fixed linear probes, no rollout); forecast merit under one shared operator, the direct multi-horizon forecaster of §3.2.1 trained identically on every state (three operator seeds; mean over the five observables of the pooled nonlinear-probe R^2 at horizon sixteen; seed means, with case-clustered bootstrap intervals in the text); field-reconstruction error (VRMSE at horizon eight); and decode similarity (SSIM). The controlled matrix (top block) moves one training ingredient at a time; the reference recipes (bottom block) use their own architectures (table 12). Wake readability separates on the wake-supervision head, not on the training objective, and under the shared operator the three wake-headed states are also the most forecastable, statistically indistinguishably from one another. The lift is read well by every anchored state and, for the predictive state, at least as well as the wake (0.813 against 0.751), so the wake supervision does not cost lift accuracy. The wake-supervised predictive state's own co-trained predictor leads every fitted operator at short horizon (0.755 at horizon eight) and is overtaken by the direct multi-horizon forecaster at horizon sixteen (0.418 against 0.561), the error-accumulation contrast of §4.4.

State	Objective	Wake head	wake R^2	lift R^2	merit	field VRMSE	SSIM
predictive (wake)	predictive	yes	0.751	0.813	0.561	0.962	0.779
supervised only	none	yes	0.792	0.769	0.443	0.992	0.774
AE (wake)	reconstruction	yes	0.750	0.765	0.574	0.976	0.776
predictive (no wake)	predictive	no	0.160	0.797	−0.426	0.983	0.767
AE (no wake)	reconstruction	no	0.456	0.825	0.285	0.976	0.773
anti-collapse AE	reconstruction	no	0.333	0.206	0.068	0.949	0.782
β-VAE	reconstruction	yes	0.728	0.668	0.040	1.007	0.724
Fukami	reconstruction	no	−0.094	0.790	−0.195	1.016	0.708
Fukami (wake)	reconstruction	yes	0.432	0.727	0.162	1.026	0.705
POD	linear	no	0.186	0.610	0.036	0.930	0.775

clustered intervals overlapping), so the forecastability of the coefficient state, like its readability, is supplied by the supervision rather than by the training objective: under the common forecaster the predictive objective does not dominate matched-supervision reconstruction in aggregate five-observable merit. What the predictive objective adds is the model it co-trains: the wake-supervised predictive state's own vector predictor forecasts the five observables at 0.755 at horizon eight, above what the shared operator extracts from any latent there (0.680), and is overtaken by the direct multi-horizon forecaster at horizon sixteen (0.418 against 0.561), the error-accumulation contrast that §4.4 quantifies. We state the representational result as the supervision's, and turn to what the predictive objective does add, which is dynamical.

One inversion in the table fixes the coordinate-level reading. The predictive latent trained without the wake head carries the highest single-observable lift forecast of any state (0.691) yet the lowest wake readability (0.160), because the forces are held redundantly across the latent while the wake is not, the distributed-code signature we return to in §4.3. Among the reference recipes the lift-augmented reconstructive autoencoder reads the wake only when it carries the wake head (0.432 against −0.094 without it), and the linear basis does not reach the wake at all (0.186), so the POD basis does not recover the enstrophy that a supervised nonlinear state does.

4.2. *Representation quality across dimension*

4.2.1. *How small the state can be*

The dimension axis splits into two tiers, and the split is the paper's compression statement (figure 5a). At $d = 4$ the predictive families read the peak loads through a linear probe at 0.903 (wake-supervised predictive state lineage) and 0.910 (coupled-objective arm) against 0.796 for the reconstructive baseline, three seeds each with non-overlapping ranges; the

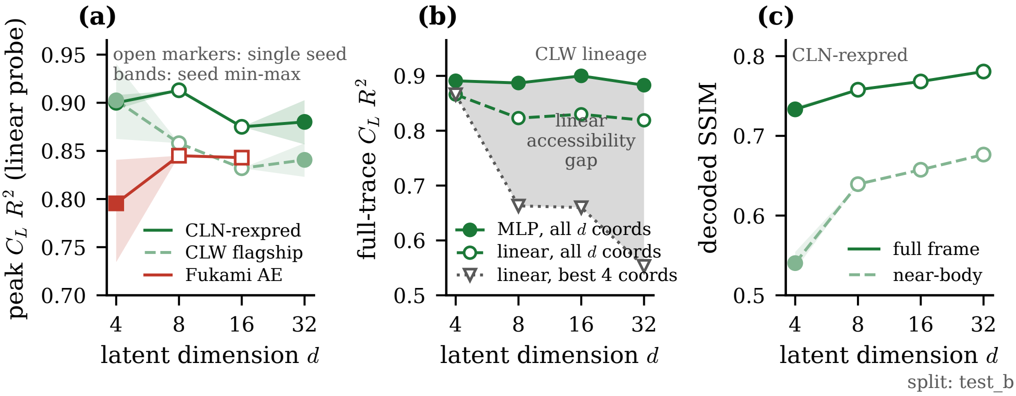


Figure 5. The dimension axis on the in-distribution test set. (a) Peak-region lift R^2 (linear probe) against latent dimension for the three lineages; shaded bands are three-seed min-max ranges, open markers single seeds. (b) The probe-dilution control: a multilayer probe is flat in d while the best-four-coordinate linear probe falls, so the $d = 4$ advantage is linear accessibility of a distributed code, not information. (c) Decoded SSIM (Wang convention, §3.5) of the same lineage rises with d: the wake tier is not compressible.

lift-focused predictive state lineage trained through the direct multi-horizon forecaster is close to dimension-insensitive on the peaks, from 0.900 at $d = 4$ to 0.880 at $d = 32$ (three seeds at both ends), where the wake-supervised predictive state lineage declines with d (from 0.902 to 0.841). A four-coefficient lift-critical state is therefore viable. The wake state is not compressible to it: the decoded structural fidelity of the same lineage rises monotonically with dimension (full-frame SSIM 0.733 to 0.781, near-body 0.540 to 0.677), so the wake-bearing state needs $d \geq 16$ while the lift saturates by $d \approx 4$ to 8.

The probe-dilution control disciplines how the first tier may be read (figure 5b). A multilayer probe nearly equalises every dimension (from 0.89 at $d = 4$ to 0.88 at $d = 32$ on the in-distribution test set): the lift information in the state is approximately unchanged over the tested dimensions, and small states do not know more. What changes with dimension is linear accessibility. The best four coordinates of a larger state recover only 0.66 to 0.55 through a linear read-out, so the code is distributed rather than diluted, and the $d = 4$ advantage is exactly that four coordinates are what a linear probe, a filter gain, or a controller reads directly. The precise form of the two-tier claim is therefore: compact and linear at $d = 4$, distributed and partly nonlinear above.

4.2.2. *The decode floor*

Decoding closes the loop on the construction story: the same division of labour that the probes read appears in the decoded fields, scored on the in-distribution test set through identical-capacity frozen decoders with the Wang SSIM convention of §3.5. With the lift anchor present, the near-body head supplies the same near-body structural fidelity as the wake head (0.709 for the lift-focused predictive state against 0.709 for the wake-supervised predictive state), the collapsed no-L cells sit far below (from 0.470 for CW), and the reconstructive anchors reach parity (0.704 for AE-LW), as an objective trained on the field should (figures 22 and the per-cell bars in the data record). The decode panels make the same point qualitatively: all three healthy states recover the near-body vortex system and the coherent wake at matched fidelity, and the residual rows concentrate on the fine-grain shed structures no 32-coefficient state carries. The decode panels are figure 22 in appendix A.2, since field decode is a wash across the compact families and supports this claim rather than leading it.

Table 3. Decode fidelity at the critical instants of the encounter, rather than averaged over the window: SSIM between each family's decoded field and the DNS truth at the pre-impact frame, the impact frame and the peak-lift frame (argmax $|C_L|$), for the full field and the near-body band, averaged over the 42 in-distribution test encounters at $d = 32$. Decode floor: the true latent is decoded, with no forecast. Both nonlinear supervised predictive states, the wake-supervised predictive state and the lift-focused predictive state, together with the wake autoencoder and the energy-optimal linear basis, all render the field faithfully at every critical instant, including the near-body band; only the published-recipe lineage is a clear step behind. Field decode at the critical instants is thus not where the compact states separate, consistent with the linear basis's field VRMSE advantage; they part on the wake observables (table 4) and on forecasting. The frozen decoders exist at a single encoder seed, so the decoded-field tables are single-seed.

	full field			near-body band		
State	pre-impact	impact	peak lift	pre-impact	impact	peak lift
predictive (wake-supervised predictive state)	0.83	0.81	0.82	0.72	0.63	0.65
predictive (lift-focused predictive state)	0.82	0.80	0.82	0.70	0.63	0.66
AE (wake)	0.82	0.81	0.81	0.73	0.65	0.64
Fukami (wake)	0.72	0.72	0.70	0.54	0.45	0.41
POD	0.82	0.82	0.81	0.74	0.67	0.67

The linear basis is not disadvantaged in field reconstruction by this comparison, and it is worth saying why before the wake result is read. By field VRMSE the proper orthogonal basis is the most accurate of the ten families (0.930, against 0.962 for the predictive state), and its decoded similarity matches the nonlinear states (0.775 against 0.779 SSIM in the Wang convention of §3.5). VRMSE rewards an energy-optimal linear basis and penalises a nonlinear state that carries a high decode floor, so we route field-fidelity claims through SSIM and the decoded snapshots and reserve the wake-observable readability for the nonlinear-content claim. There the linear basis reaches only 0.186 at $d = 32$, and the POD coefficients do not recover the wake at any tested dimension. The proper orthogonal basis holds the shedding clock (§4.3) and stays competitive on the forces; the finding is not that it is obsolete but that the POD coefficients do not achieve comparable linear readability of the wake-enstrophy target at the dimensions tested.

The window-averaged decode floor can be resolved to the critical instants of the encounter, the pre-impact frame, the impact and the peak lift (table 3). Through the impact the compact nonlinear supervised states and the linear basis all render the field faithfully, at SSIM 0.81 for the predictive state, 0.81 for the matched-supervision reconstructive and 0.82 for the linear basis (near-body 0.63, 0.65 and 0.67), with only the published-recipe lineage a clear step behind (0.72 full, 0.45 near-body). Field decode at the critical instants is therefore not where the compact states separate: the energy-optimal linear basis reconstructs the instant as sharply as the nonlinear states, the same VRMSE advantage read above, and the families part company instead on the wake observables, which the POD coefficients do not reach, and on forecasting.

Across dimension the picture is the same (figure 23 in appendix A.2): decode fidelity at the impact instant rises and saturates for the compact nonlinear states and the energy-optimal linear basis together, the two indistinguishable at every dimension, while the published-recipe lineage stays lower. Field decode does not separate the compact states at any dimension.

The observables tell the discriminating story the field pixels hide (table 4). A linear probe reads the lift and the wake enstrophy off each state at the same three instants. Supervision decides what a state carries: the wake-supervised predictive state and the wake autoencoder carry both the lift and the wake through the transient, whereas the lift-focused predictive state reads the lift as accurately as any state ($R^2 = 0.78$ at impact) but not the wake

Table 4. Where the families separate at the critical instants: held-out readability of the lift C_L and the wake enstrophy E_w at the pre-impact, impact and peak-lift instants (±2 frames), each cell the coefficient of determination R^2 and, in parentheses, the physical error (RMSE, in lift-coefficient units for C_L and in the pipeline's enstrophy units for E_w, whose test-window standard deviation is 92). A linear probe fit on the training window reads each observable off the frozen latent; $d = 32$, on the 42 in-distribution test encounters. The learned predictive and AE states are means over three encoder seeds (encoder-seed standard deviation ≤ 0.09 in R^2 on every cell that reads the quantity); the published-recipe lineage has a single seed at this dimension and the linear basis is deterministic. Unlike the decoded field (table 3), the observables separate the states exactly where it matters. The wake-supervised predictive state and the wake autoencoder read both the lift and the wake enstrophy through the transient; the lift-focused predictive state reads the lift as accurately as any state ($R^2 = 0.78$ at impact) yet reads the wake enstrophy far worse (near zero at impact, 0.05), locating the wake content in the wake head rather than the lift anchor. The linear basis loses the lift at the impact instant ($R^2 = -0.05$) and carries the largest lift error at every instant, including where its R^2 looks healthy at the high-variance peak (0.74 but RMSE 1.80, the worst in the column), and it never reads the wake enstrophy. The energy-optimal linear basis renders the field but cannot read the load or the wake at the moments that decide the encounter.

	lift C_L: R^2 (RMSE)			wake enstrophy E_w: R^2 (RMSE)		
State	pre-impact	impact	peak lift	pre-impact	impact	peak lift
predictive (wake-supervised predictive state)	0.93 (1.00)	0.70 (0.84)	0.83 (1.44)	0.61 (59)	0.56 (59)	0.87 (46)
predictive (lift-focused predictive state)	0.93 (0.98)	0.78 (0.71)	0.84 (1.38)	−0.82 (127)	0.05 (85)	0.42 (96)
AE (wake)	0.89 (1.22)	0.76 (0.75)	0.81 (1.53)	0.39 (74)	0.70 (48)	0.83 (52)
Fukami (wake)	0.93 (0.98)	0.79 (0.69)	0.81 (1.55)	0.74 (48)	0.72 (48)	0.58 (81)
POD	0.77 (1.74)	−0.05 (1.57)	0.74 (1.80)	0.66 (56)	0.31 (74)	0.31 (105)

(0.05, against 0.56 for the wake-supervised predictive state), so the wake content is the wake head's contribution and not a by-product of the lift anchor. The linear basis fails differently again, losing the lift at the very impact instant ($R^2 = -0.05$, against 0.70 for the predictive state) and never reading the wake enstrophy (0.31 through impact). It is at the critical moments, not on the window average, that the linear readout breaks down: the energy-optimal basis renders the field but cannot read the load or the wake exactly when the encounter turns nonlinear.

4.2.3. *The distributed wake code*

The wake is a collective code rather than a labelled coordinate. The gap between the wake readability of the full predictive latent and that of its single most informative coordinate is 0.51, and 0.53 for the objective-free supervised state, against −0.03 and 0.04 for the reconstructive and linear baselines, so no single predictive coordinate carries the wake and the information is spread across many. The dimension at which each family first reaches a wake R^2 of one half makes the same point from the other side: the predictive and the wake-headed reconstructive states need eight latent dimensions (8 each), while the linear basis never reaches it at any dimension we tested. The headline dimension of thirty-two sits on a plateau rather than a cliff, the wake readability moving only 0.053 across sixteen, thirty-two and sixty-four (0.751 at thirty-two), so the choice of dimension is a robustness statement and not a tuned operating point.

A spatially resolved latent decodes the vorticity field more sharply than this coefficient state, but it is of far higher dimension than the wall measurement and so is not recoverable from sparse pressure. That decodability-versus-estimability trade is the reason the coefficient state, not the field-resolved latent, is the object this paper estimates; we take it up in §5.3 and quantify its cost in the supplementary material.

4.3. *The physics the latent state holds*

Before the state is forecast or estimated, we read what the representation holds, in measures that do not depend on any filter. The predictive latent carries the shedding clock. Fitting a linear operator to the encoded limit-cycle trajectory by dynamic mode decomposition places a marginally stable oscillatory mode at Strouhal number 0.666, against the measured shedding line, with modulus 0.993 just below unity. The Fukami-lineage autoencoders lose that clock: their leading latent mode is heavily damped and off-frequency (0.182 at modulus 0.590 without the wake head, 0.147 at 0.294 with it). The structured states keep it, whether their structure comes from prediction, from supervision, from anti-collapse, or from a linear basis: the objective-free supervised latent, the anti-collapse reconstruction, and the proper orthogonal basis all place the shedding mode near the measured frequency on a near-unit-modulus orbit (0.654, 0.676 and 0.673 at moduli 0.989, 0.995 and 0.996). It is the Fukami-lineage reconstruction, not reconstruction in general, that fails to hold the shedding spectrum.

The coordinate spectra say the same in the frequency domain and locate the reconstruction failure precisely. Averaging a per-coordinate Welch spectrum over the training encounters, the published-recipe reconstruction spreads its latent energy across frequency, at a median spectral flatness of 0.068 against 0.012 for the predictive latent, 0.024 for the matched-supervision reconstruction and 0.030 for the linear basis (figure 6, bottom); the published recipe exceeds the predictive latent by 0.056 in flatness with a case-clustered interval [0.048, 0.065] that excludes zero. The predictive coordinates are the most sharply peaked at the shedding line, and it is the published-recipe reconstruction alone, not the matched-supervision autoencoder, whose coordinates are broadband: the broadband, phase-scrambling latent that a reconstruction objective without matched supervision produces is the frequency-domain signature of the divergence the same latents show under rollout. The same contrast appears in the phase portraits of the latent trajectories (figure 16): the predictive, linear and matched-supervision states trace coherent cycles that the gust perturbs, while the published-recipe reconstruction wanders without a clean orbit.

The parametric manifold coheres with the release parameters where the training data resolve them. An impact-frame probe of the predictive latent recovers the gust strength and the core diameter well (0.88 and 0.73 on the in-distribution test set), which is the atlas of figure 6 (top) coloured by parameter. The wall-normal offset, unreadable at the linear-probe level in earlier work, becomes weakly readable here (0.29 linearly and 0.79 with a nonlinear probe), plausibly because the additional cases sample it more densely near the training centre; we note this rather than lean on it, since the offset remains the least resolved parameter (§2.2).

A state that reads the wake at a single instant need not survive a rollout, because a probe attached to a drifted latent is queried outside its support. We ask where each rolled state goes relative to the training distribution by decomposing its departure onto the principal and the near-null directions of the encoded covariance (table 14). The reconstruction carrying only an anti-collapse term is near-isotropic, its covariance condition number close to unity (11), and its rollout leaks a tenth of its departure into the near-null directions (0.100). The supervised states, predictive and objective-free alike, are strongly anisotropic, their variance concentrated in a low-rank subspace (condition numbers 451 and 1041), and their rollouts keep their departures inside it (0.014 and 0.029, against the 0.25 of an isotropic null). Supervision, not the anti-collapse term alone, builds the protected subspace: it pins the encoded variance onto the observable-aligned directions, and the forward model, having only ever seen states there, moves within it. The anti-collapse term on its own drives the latent toward isotropy, as an independent adaptation of the same regulariser to biosignals also reports (Broustail *et al.* 2026), and it is the supervision that supplies the geometry the rollout needs.

4.4. *Forecastability*

The family ordering survives an operator-robust protocol, which resolves the operator confound the suited-operator comparison of §3.2 left open: one direct multi-horizon forecaster (§3.2.1), trained identically on each family's frozen latents, is stable on every latent including the reconstructive ones. Under it the predictive states carry the more forecastable lift, decoded lift $R^2 = 0.638$ for the wake-supervised predictive state and 0.685 for the lift-focused predictive state against 0.539 for the reconstructive anchor, three operator-training seeds per family on the in-distribution test set, with the lift-focused predictive state latents the most lift-forecastable of all (figure 7a). On the five-observable merit of table 2, recomputed under the same shared operator across all ten families, the wake-headed states are statistically tied (§4.1), so the operator-robust statement is two-fold: supervision, not the training objective, decides which states are forecastable at all, and the lift-focused predictive state leads specifically on the lift readout.

The forecasting mechanism is the direct-versus-autoregressive contrast. Rolled autoregressively through the impact, the matched transformer compounds its own errors: forty steps from the pre-impact context its decoded lift closes at -0.62 on the wake-supervised predictive state latents and -0.23 on the lift-focused predictive state latents. The direct multi-horizon forecaster, emitting all forty steps in one shot from the same context, closes at 0.701 on the identical protocol (figure 7b): the large improvement of the direct multi-horizon forecast indicates that recursive error accumulation is the dominant limitation of the autoregressive implementation over this horizon. The direct forecast is a conditional median, and the correct reading is that of a median: it contracts toward the climatology of the shedding cycle where the encounter is unresolved, which is precisely the model-error regime the assimilation of §4.6.3 must correct.

Knowing the gust does not help. Conditioning the forecaster on the true episode parameters (G, D, Y), an oracle no deployment has, degrades the forecast in every one of three operator seeds, 0.539 against 0.699 unconditioned, with non-overlapping seed bands: at the present case count the oracle covariates overfit held-out parameter combinations. A deployable phase covariate is a wash within seed noise (0.710). This closes the estimation thesis from the third side: forecast alone contracts, parameter oracles hurt, and what remains is the model-plus-pressure route of part D.

The forecast is not uniformly hard across the encounter, and resolving it by phase locates where the families separate (figure 24). Splitting every shared-operator forecast by whether its horizon stays in the pre-impact shedding or crosses the impact transient, merit is highest before impact and drops through it for every family, the wake-supervised predictive state from 0.78 pre-impact to 0.63 through impact at horizon eight. The separation is in the transient: through the impact the wake-supervised states carry the encounter (0.63 for the wake-supervised predictive state and 0.70 for the reconstructive AE-LW at horizon eight, 0.57 and 0.61 at horizon sixteen), while the published-recipe reconstruction collapses across it (near zero at horizon eight, -0.29 at horizon sixteen) and the linear basis holds the shedding but fades into the relaxation (through-impact 0.36 against post-impact -0.16 at horizon sixteen). The through-impact window, not the periodic shedding, is what a forecast model must survive, and it is where the anchored, wake-supervised states hold while the reconstruction fails (figure 24 in appendix A.2).

When the forecast is decoded to the field rather than read in the latent, the families no longer separate. Decoding each family's forecast through its frozen decoder and scoring SSIM against the DNS truth, all three states forecast the field to comparable fidelity through

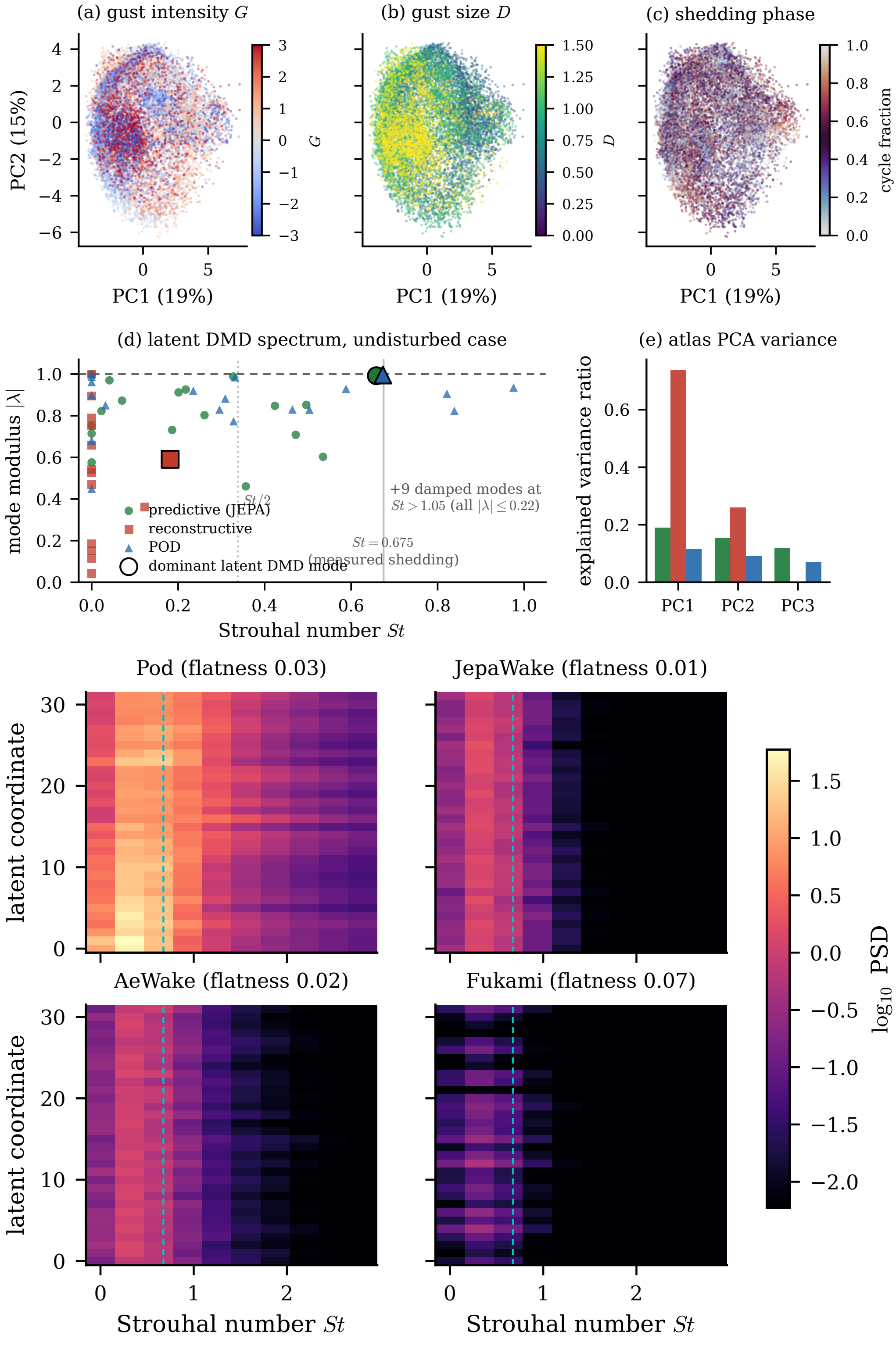
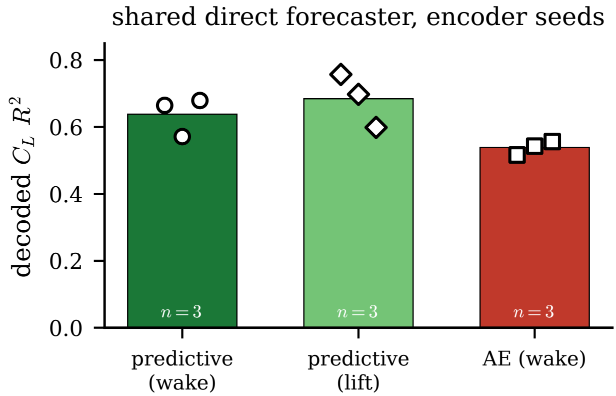
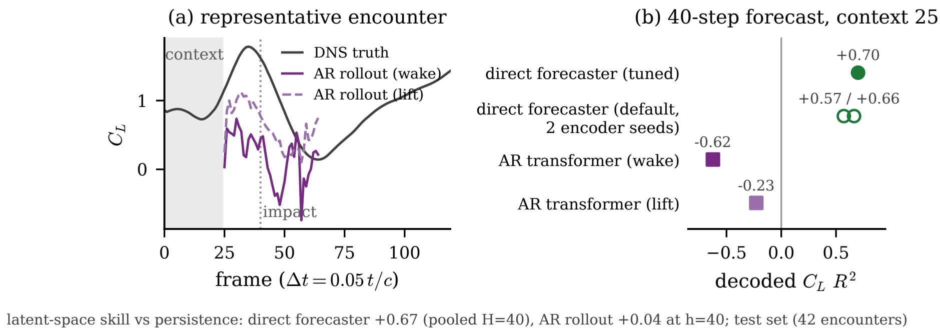


Figure 6. The physics the latent state holds. Top: two-dimensional projections of the predictive coefficient latent coloured by gust strength, core diameter and shedding phase, showing a parametric manifold that coheres with the release parameters, with the latent dynamic-mode-decomposition spectrum (modulus against Strouhal number, by family; the predictive and linear latents place a marginally stable mode at the measured shedding frequency while the Fukami-lineage reconstruction is damped and off-frequency). Bottom: per-coordinate latent spectra by family (training encounters, d = 32; mean Welch power spectral density against Strouhal



(a)



(b)

Figure 7. Forecasting the state on the in-distribution test set. (a) Decoded lift R^2 of the shared direct multi-horizon forecaster per family (three operator seeds each, bars at the seed mean): the wake-supervised states forecast the lift best. (b) Direct versus autoregressive over forty steps through the impact from a 25-frame context: the one-shot forecast stays usable where the rollout compounds. The conditioning null, where the (G, D, Y) oracle degrades the forecast and the phase covariate is a wash, is figure 18 in appendix A.

the impact at horizon eight, SSIM 0.78 for the predictive state, 0.76 for the reconstructive and 0.76 for the linear, the predictive state only modestly ahead on the near-body band (0.54 against 0.52 and 0.49). The decoded field, like the decode floor of §4.2, is not where the states separate: the energy-optimal linear basis renders the forecast field about as sharply as the nonlinear ones. What separates them is the observable content the field-averaged SSIM washes out, the wake-observable merit above, where the wake-supervised and predictive states lead and the linear basis does not.

4.4.1. *Why the state stays usable under rollout*

What the predictive objective adds is long-horizon stability, and it is the multi-step form of the objective that adds it. Trained with a single-step rather than a multi-step rollout, the same predictive latent forecasts the five observables less well at every horizon, and the margin widens with the rollout: by sixteen steps the single-step latent's merit has fallen to 0.237 while the multi-step latent holds 0.488, with markedly lower on-manifold drift (0.518 against 0.733 at horizon sixteen). The single-step latent is marginally better on the load alone a few steps out (0.685 against 0.664 in C_L closure at eight steps), a single-observable exception to a forecast advantage that otherwise grows with horizon. The objective-free supervised state, which has the readability and the protected subspace but no predictive

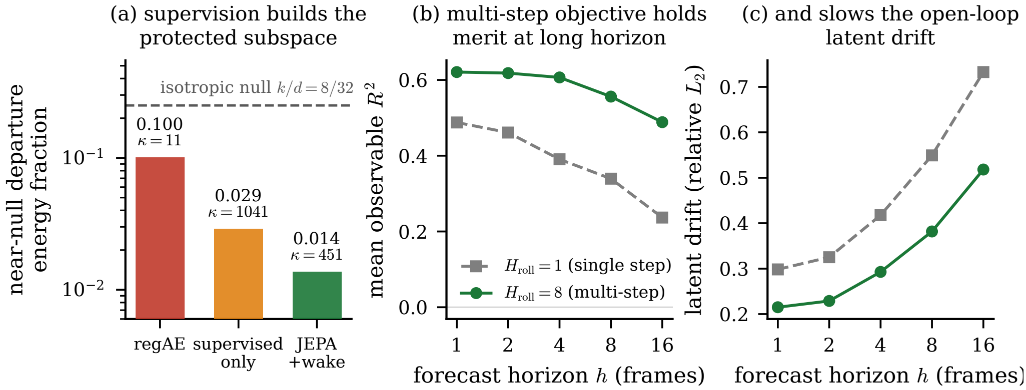


Figure 8. The rollout mechanism and what the multi-step objective adds. (a) Near-null departure fraction by family against the isotropic null, annotated with each latent covariance condition number: supervision builds the protected subspace. (b) Forecast merit and (c) rollout drift against horizon for the single-step and multi-step objectives, tied at one step and separated by sixteen.

training, is the weakest field forecaster of the supervised group. The contribution of the predictive objective is therefore forward rather than instantaneous: it keeps the rolled state both in distribution and near the true trajectory, which is exactly what a sequential estimator queries.

4.5. *What wall pressure observes*

At deployment the state must be read from the wall, so we ask how much of each latent a sparse set of pressure taps recovers over a window ending at the readout instant (table 5). Two findings organise the answer. First, raw-variance recoverability and physics recoverability disagree: the reconstruction latent is the most recoverable in state variance (0.921) yet carries the least wall-recoverable physics (0.470 in observable closure), while the predictive latent carries the most physics (0.589) from a less recoverable state (0.659 ± 0.009 over three seeds). The wall aligns with many of the reconstruction latent's directions, but they are the low-variance ones that do not carry the observables; it aligns with fewer of the predictive latent's directions, but those are the ones that hold the state. Second, perturbing the predictive state along its principal directions and measuring the signal that reaches the pressure head shows the wall reading the force and wake-enstrophy directions and blind to the highest-variance wake-circulation direction, a ranking unchanged under a fixed-amplitude perturbation and so a property of the learned dynamics rather than of the normalisation. The wall, in short, reads the force well and the wake circulation not at all, which is why the state must be estimated sequentially rather than inverted from one frame.

The margin of the coefficient state over a raw high-dimensional field latent is real but modest. Its recoverability advantage is 0.120 in R^2, with a confidence interval $[0.096, 0.145]$ that clears zero. We therefore do not claim a strong-form estimability gate; the defensible statement is the clean physics-versus-variance ordering above.

4.5.1. *Sensors traded for delays*

That the wall is blind to the wake circulation at a single instant is a statement about one frame, not about the sensor count, and the delay-embedding reading of §3.3 makes a testable prediction: the circulation the instantaneous pressure misses should become recoverable as the pressure is read over a longer window. It does. Holding the tap count at eight and lengthening the window, the recovery of the wall-blind wake circulation rises from near its

Table 5. What eight wall-pressure taps recover of each family's coefficient state, on the in-distribution test set, with the recovery estimator selected by cross-validation per family and the observables read through the frozen probes. Raw-variance recoverability and physics recoverability disagree: the reconstruction latent is the most recoverable in state variance but the least in physics, while the predictive latent carries the most wall-recoverable physics from a less recoverable state. The absolute state recovery is estimator-limited: a per-family tuned LSTM (hidden width validation-selected up to 512) recovers more from the same taps, raising the predictive state to 0.833 and the reconstruction to 0.937, and it had not saturated at the largest network tested, so these are a lower bound on the wall-recoverable state; the family ordering, which is the claim, is unchanged.

State	recovery estimator	state R^2	observable R^2
predictive	LSTM	0.659	0.589
reconstructive (Fukami)	LSTM	0.921	0.470
linear (POD)	MLP	0.561	0.534

single-frame value to 0.33 at a thirty-frame window, a gain of 0.224 with a case-clustered interval $[0.136, 0.343]$ that excludes zero, while the force, already visible at one instant, barely moves. The delays recover the coordinate the sensors cannot.

Varying the tap count and the window together traces a spatial-for-temporal trade (figure 9). A single tap read over a thirty-frame window approaches the recovery obtained from eight taps read at one instant, and intermediate combinations of tap count and window meet in between, because the encounter is low-dimensional and a state that filled its ambient dimension would not admit the trade. The eight-tap, thirty-frame recovery reconciles exactly with the static recovery above (0.656), and the delay stride sits at the cache cadence, close to the first mutual-information minimum of the tap signals at 29 frames.

The grid uses a shared target-blind tap array so that the trade is free of any model-conditioned placement; rerunning its eight-tap, thirty-frame cell on the model-conditioned taps of table 5 reproduces that table's recovery exactly (0.656 against 0.659), so the two analyses meet where their sensing coincides. This trade belongs to the static delay-coordinate reconstruction, and it does not transfer to the frozen sequential filter, which needs its eight taps. Rerunning the filter with only its own leading taps, and changing nothing else, drives the analysis lift below the skill of the encounter mean at every gust strength the eight-tap filter tracks: at $|G| = 1$ the single-tap analysis is worse than the eight-tap filter by a paired margin whose interval $[-107.96, -4.78]$ excludes zero, and the loss stays significant at $|G| = 1.5$ and $|G| = 2$. The rank of the innovation, with the process and observation noise held at their calibrated values, is what the filter cannot spare: its sensor budget is a calibration constraint, not a delay-embedding one, which points at the noise model of §5.4 rather than at fewer sensors.

4.6. *Estimating the encounter from the wall*

4.6.1. *Tracking the encounter from the wall*

We now run the estimator. It is initialised from wall pressure alone before impact and corrected at every frame through the encounter, and it never sees the vorticity field. Figure 10 follows representative held-out encounters, the true lift and wake enstrophy against the open-loop rollout with no correction, the sliding-window pressure-only regression with no dynamics, and the filter analysis. The open-loop rollout tracks the pre-impact baseline and then loses the encounter, as a forecast with no measurement must; the pressure-only regression follows the force while its window is representative and goes stale through the transient; the filter follows the load through impact and into relaxation. This realises, on the

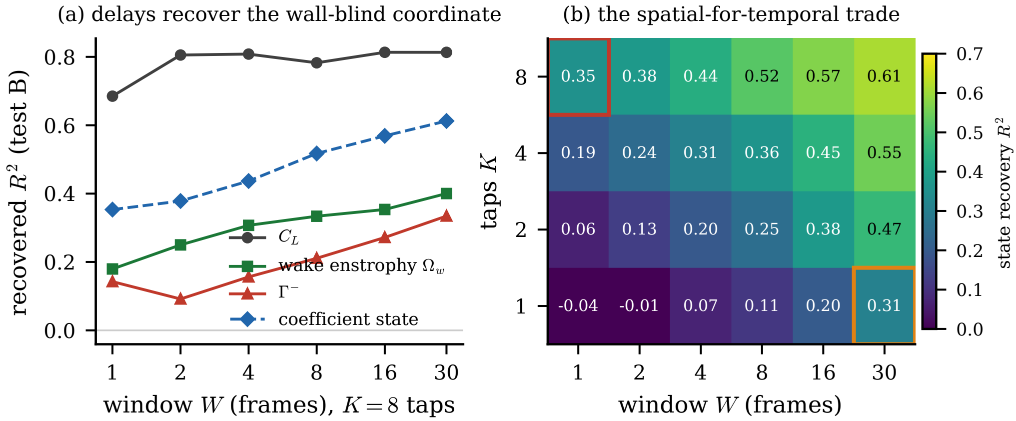


Figure 9. Sensors traded for delays. (a) Recovery of each observable from eight wall-pressure taps against the length of the pressure window (in-distribution test set); the wall-blind wake circulation rises with the window while the force, visible at one instant, does not. (b) Coefficient-state recovery over the tap-count by window-length grid, with the single-tap, thirty-frame cell and the reference eight-tap, single-frame cell outlined: a long window with few taps matches a short window with many.

wake-supervised coefficient state, the online-estimation pathway the extreme-aerodynamic control literature has named but not closed (Fukami *et al.* 2024; Mousavi & Eldredge 2025; Eldredge & Mousavi 2025): the state a real-time controller would need is recovered from sparse wall pressure and advanced by the learned dynamics, without ever measuring the field.

Two boundaries frame what this demonstration does and does not establish, and we state them before the envelope. First, tracking the load inside the training range is not unique to this state. Running the identical frozen filter on every family (table 6), the nonlinear latents all track the lift for $|G|$ between one and two, and the wake-supervised reconstructive control is the best of them at the mildest gusts (0.86 at $|G| = 1$ against 0.74 for the predictive state); only the linear basis fails outright. What no static approach achieves is the strong-gust range: a windowed regression from the same taps straight to the lift, with no latent at all, fades to 0.08 by a gust ratio of three, and even the full wall array recovers only 0.14 at the boundary, so the sequential accumulation of pressure through a learned nonlinear state, rather than any particular training objective, is what carries the load into the strong-gust regime. Where the objectives separate is at that boundary and in the filter's statistical health. The predictive state combines a boundary closure among the highest (0.84, against 0.66 to 0.85 for the alternatives), by far the lowest consistency-failure rate there (0.72 against 0.93 to 0.97), consistency across gust strengths where the wake-headed reconstructive baseline posts −0.39 at $|G| = 1$, and the least-degraded wake readout. Second, no family's filter tracks the wake enstrophy: the analysis wake closure is below the mean predictor for every state, least badly for the predictive one, so the wake content of the encounter is established at the representational tier (§4.1.2) and is not yet recoverable through the assimilation. The filter is a load estimator whose state carries wake information; it is not a wake estimator, and we say so plainly.

How far into the gust the load recovery holds is the operating envelope of §4.6.2.

4.6.2. *The gust-intensity operating envelope*

This subsection diagnoses the reach of the base frozen filter across the envelope; the production estimator whose performance we report is the two-stage filter of §4.6.3, and the

Table 6. The frozen filter run identically on every family: median analysis lift closure by gust strength, with the consistency-failure rate (§3.3.2) at the $|G| = 4$ boundary, and the direct no-latent baseline (windowed regression from the same eight taps straight to the lift) beneath. Load-tracking inside the training range is a property of the nonlinear latents generally, not of any one objective; the families separate at the strongest gusts, in calibration, and in what else the tracked state carries (§4.6.1). The linear basis and the no-latent regression both fail beyond $|G| = 2$.

state	$\|G\| = 1$	1.5	2	3	4	cons. fail. at 4
predictive (wake)	0.74	0.83	0.73	0.63	0.84	0.72
AE (wake)	0.86	0.86	0.81	0.49	0.66	0.93
Fukami	0.79	0.87	0.78	0.68	0.76	0.97
Fukami (wake)	−0.39	0.72	0.68	0.77	0.85	0.95
POD	0.53	0.52	−0.33	−0.27	0.25	0.93
direct regression (no latent)	0.41	0.29	0.21	0.08	0.08	–

Table 7. The operating envelope, by gust strength, for the predictive coefficient state under the base frozen filter, used here as a diagnostic; the final production estimator is the two-stage filter of §4.6.3. Median analysis lift closure, median analysis lift error in lift units, and consistency-failure rate of the filter (the fixed innovation-consistency criterion of §3.3.2, not a load-tracking failure) against the static single-frame recovery, over all encounters (train, validation, in-distribution test, and the $|G| = 4$ boundary test sets), stratified by $|G|$. The $|G| \leq 0.5$ closure is near zero because the true lift is near-constant there and the R^2 denominator vanishes, not because the filter fails; the per-encounter paired tests on the in-distribution test and boundary test sets are in Appendix A.

$\|G\|$	filter C_L R^2	filter C_L RMSE	consistency-failure rate	static recovery C_L R^2
≤ 0.5	0.25	0.23	0.00	0.34
1	0.74	0.31	0.00	0.73
1.5	0.83	0.38	0.09	0.65
2	0.73	0.63	0.33	0.24
3	0.63	0.68	0.68	−0.46
4	0.84	0.72	0.72	−0.33

two are kept distinct throughout. A reduced state earns its place at deployment only if it can be followed from the sensors that are present, and the wall pressure a small vehicle can carry does not, at a single instant, fix the wake it leaves behind (§4.5). The question is therefore not whether the state is readable but how far into the gust an estimator that sees only the wall can track it. The sequential filter reaches much further than the single-frame inverse. While the encounter stays mild, the static single-frame recovery of the lift holds, with a held-out R^2 of 0.73 at $|G| = 1$ and 0.65 at $|G| = 1.5$; as the gust strengthens it collapses, falling below the skill of predicting the encounter mean by a gust ratio of three (−0.46) and remaining there at the $|G| = 4$ boundary (−0.33), because a single pressure frame cannot see the reorganised wake a strong gust deposits. Reading the same taps through the learned dynamics, the filter follows the load further, but the accuracy that matters at deployment is the error in real lift units, not the variance-normalised score, and that error grows with the gust. The median analysis C_L error rises from 0.31 at $|G| = 1$ to 0.68 at a ratio of three and 0.72 at the boundary (table 7), even as its R^2 rises to 0.84: the strong gust inflates the lift excursion faster than the error grows, so the normalised score flatters the extreme-gust tracking while the absolute error roughly doubles. The open-loop forecast never tracks the load at any gust strength, so it is the correction step, not the forward model, that recovers the state, and it does so from pressure the visibility analysis found blind to the wake circulation.

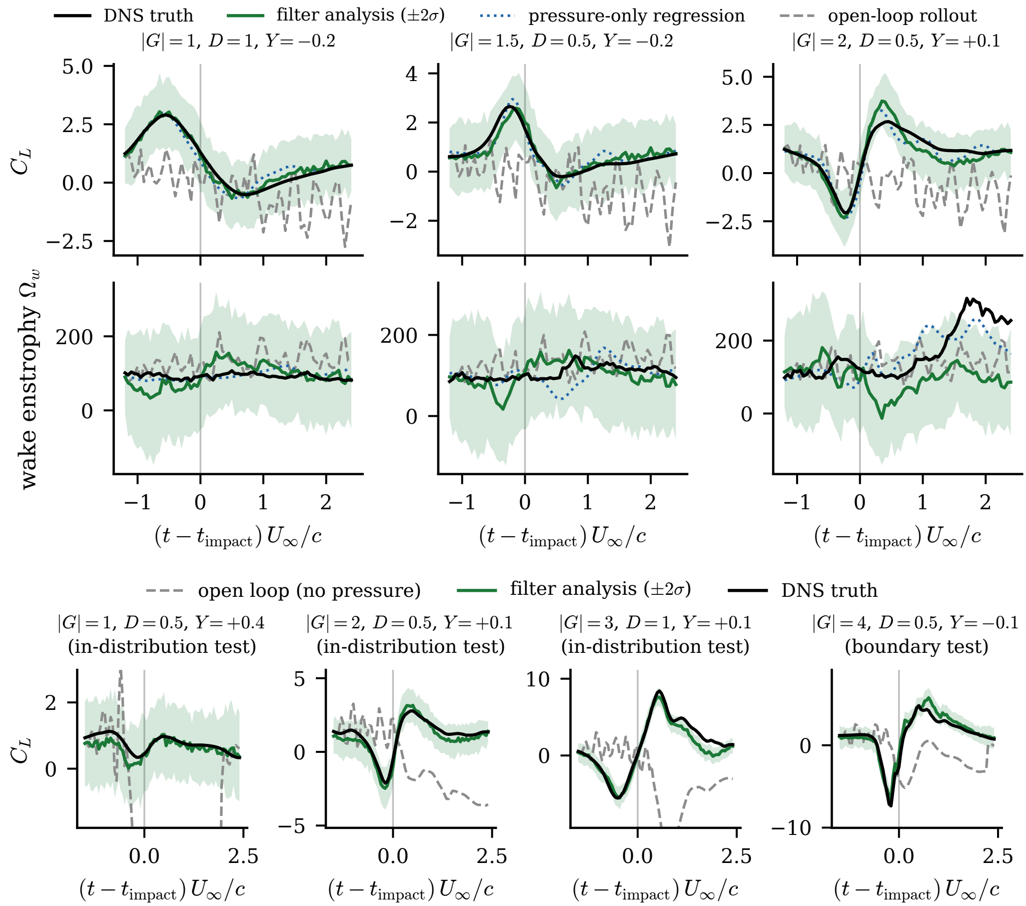


Figure 10. The estimator tracks the encounter from wall pressure alone. Top: representative held-out encounters at $|G| = 1$, 1.5 and 2 (columns), lift coefficient and wake enstrophy (rows); each panel shows the DNS truth, the filter analysis with its $\pm 2\sigma$ ensemble band, the pressure-only regression, and the open-loop rollout, the vertical line marking impact. The filter follows the load through impact and relaxation where the open-loop forecast loses it and the pressure-only regression goes stale. Bottom: C_L estimation across the gust-intensity envelope on the wake-supervised predictive state ($d = 32$) under the two-stage filter of §3.3.3 (one representative encounter per $|G|$ stratum, in-distribution test set for $|G| \leq 3$ and boundary test set at $|G| = 4$, where this filter posts no large-error encounter); inside the impact window the members are advanced by the direct multi-horizon forecaster's median with its quantile band as state-dependent process noise (production scale $c = 1.77$, validation-calibrated) and corrected by the encoded pressure observation, outside it by the matched transformer with the tap-space update. Eight wall-pressure taps, $N = 64$ members; the lift probe producing the read-out never enters the innovation; grey is the same forecast run open loop.

The lower panels of the tracking figure (figure 10) show the two-stage filter of §3.3.3 at work, one representative encounter per gust intensity: the analysis follows the lift from the pre-impact shedding through the impact excursion and the relaxation at every intensity, including the $|G| = 4$ boundary, while the same stage-scheduled forecast models run open loop, with no pressure, lose the transient.

Run identically on each family, each under its own validation-tuned inflation, the comparison is a division of degradation rather than a clean win (table 8). At moderate gusts every filter tracks the load, the reconstruction filter marginally best. At the $|G| = 4$ boundary every filter is stressed, its innovations flagged inconsistent on 0.72 to 0.95 of

Table 8. What the wall-pressure filter costs in real lift units. Median analysis C_L error (RMSE, dimensionless) and filter consistency-failure rate at the impact window, by gust strength, each family run with its own validation-selected inflation (ρ = 1.00 for the predictive filter, 1.05 for Fukami, 1.00 for POD). The consistency-failure rate is the fraction of encounters flagged by the fixed innovation-consistency criterion of §3.3.2 (sustained normalised innovation squared or an analysis Mahalanobis ratio above the threshold); the ensemble mean can still track the load where the consistency diagnostic fails. The variance-normalised R^2 rises with gust amplitude and hides the cost: at $|G| = 4$ the predictive filter scores $R^2 = 0.84$ while its absolute error is 0.72. Two readings follow. At moderate gusts every filter tracks the load, the reconstruction filter marginally best. At the strongest gusts every filter degrades, with consistency-failure rates of 0.72–0.93; the predictive filter degrades least, in both error and consistency, and the linear filter worst. The strongest gusts remain a degraded regime for every filter, not a solved one.

	median C_L RMSE			consistency-failure rate		
Filter	$\|G\|$=1	$\|G\|$=2	$\|G\|$=4	$\|G\|$=1	$\|G\|$=2	$\|G\|$=4
predictive	0.31	0.63	0.72	0.00	0.33	0.72
reconstructive (Fukami)	0.26	0.61	0.79	0.02	0.53	0.95
linear (POD)	0.36	1.14	1.33	0.01	0.49	0.93

encounters; the predictive filter degrades least, with the lowest median error (0.72 against 0.79 for the reconstruction and 1.33 for the linear basis) and the lowest consistency-failure rate, and the linear filter degrades worst. The predictive state's advantage at the boundary is one of robustness, a degraded regime rather than a solved one.

What weakens with gust strength is not the tracking, whose median holds through $|G| = 4$ (table 7), but the filter's own account of its uncertainty, which degrades from a gust ratio of about three. The consistency-failure rate climbs from none at $|G| = 1$ to a fraction 0.72 of encounters at the boundary, and the mean normalised innovation squared from 0.6 in the undisturbed limit to 15.2 at $|G| = 4$, crossing the half-consistency-failure mark at a gust ratio of 3.0 for the intermediate and the widest cores alike, at the resolution of the tested $|G|$ grid; the widest core already reaches a rate of one half at $|G| = 2$, and the most compact does not cross the mark inside the envelope. The ensemble collapses while the analysis mean still follows the load, so the filter is under-dispersed rather than mistracking. We read this as the signature of a missing noise model: the innovations are correlated in time rather than white, and the remedy is a calibrated process and observation covariance rather than the covariance inflation we deliberately withheld (§5.4).

The envelope is not symmetric in the sign of the gust, and the asymmetry survives in the linear basis, so it belongs to the flow rather than to the representation (figure 11(b)). The physically positive gusts, which roll up the larger and longer-lived leading-edge vortex (§4.3), leave the wider usable envelope. Sampling the $|G| = 4$ boundary at both signs (§2.2) is what lets this asymmetry be measured rather than assumed.

The boundary sits at a gust ratio of three, inside the training range, so it marks a limit of what the mid-plane pressure can observe and the filter assimilate, not a gap in training coverage. Read through delay coordinates the two limits are one. The filter follows the load where the single frame fails because it reconstructs the state from a window of pressure rather than an instant (§3.3), and it loses its calibration where the strengthening gust drives the interaction three-dimensional and the mid-plane pressure history captures a smaller fraction of the state (§4.5). Only at $|G| = 4$ does the mid-plane representation meet the substantially larger three-dimensional content the field itself carries (§2.1), now seen from the filter. The envelope is therefore two claims with different owners: its extension beyond every static delay-embedded inverse belongs to the sequential estimator on a nonlinear latent, a property the family comparison of §4.6.1 shows is shared; its width at the strongest

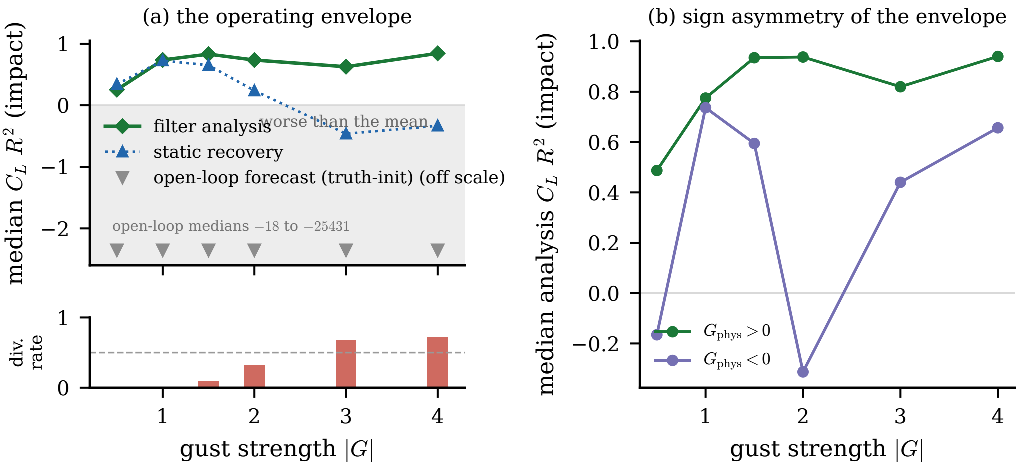


Figure 11. The gust-intensity operating envelope, diagnosed with the base frozen filter (the production estimator is the two-stage filter of §4.6.3, figure 12). (a) Median analysis lift closure (R^2, impact window) against gust strength for the base filter, the static single-frame recovery, and the truth-initialised open-loop forecast, with the filter consistency-failure rate as the strip beneath; the shaded band marks closure worse than predicting the encounter mean, and the open-loop forecast lies off the scale below it. The base filter tracks the median throughout but becomes under-dispersed from a gust ratio of about three, where the consistency-failure rate climbs. (b) The same base-filter closure resolved by the physical sign of the gust, showing the wider usable envelope of the positive gusts.

gusts, its calibration margin, and the wake content of the state being tracked belong to the wake-supervised predictive state. This envelope, not the single-instant readability, is what a wall-estimable reduced state is for.

4.6.3. *The estimator ladder, its calibration and its deployment envelope*

Physical units reorganise the phase story. Read in R^2, the relaxation phase looks like a failure (-1.17 for the filter on the in-distribution test set); read in lift units it is the best phase the estimator has (root-mean-square error 0.18 against 0.30 at impact and 0.29 before it), because the relaxation variance is small and the ratio, not the error, collapses (figure 12). Assimilation is decisive at impact: the open loop drifts to 1.54 where the filter holds 0.30, a five-fold reduction, with the median peak error at 11.7 per cent of the true peak and peak timing within one frame. Relative accuracy is scale-invariant across the training envelope, near 12 to 14 per cent at every gust intensity for the forecast-derived filter while the linear filter is non-uniform (figure 27); the absolute error grows with the peak it tracks, which is what a scale-invariant estimator must do.

The estimator ladder orders the suite at impact (appendix C, figure 29), and its top rung is unexpected: the static delay-embedded inverse, no dynamics at all, posts the best in-distribution test impact median (0.83). The filters' case does not rest on the in-distribution median. It rests on the tails, where the static delay-embedded inverse posts the ladder's single large-error encounter, its impact-phase R^2 falling below -1; on the extrapolation boundary, where the two-stage filter reaches 0.837 median R^2 on the $|G| = 4$ boundary test encounters with no large-error case and a positive relaxation (0.120); and on state tracking beyond the single read-out the inverse regresses. The linear filter with encoded observations adds the robustness column: zero consistency failures under the fixed criterion of §3.3.2 and graceful degradation as the assimilation rate drops, where the transformer-forecast hybrid collapses already at every-second-frame updates (figure in appendix B).

Calibrating the forecast-derived noise did not go the way the calibration hypothesis expected, and the failure is informative. On the held-out training encounters the pooled impact-phase innovation statistic sits below one at every band scale in the grid fixed in advance (0.53 at the selected edge $c^* = 1.00$) while the same split's tracking improves monotonically with the band (figure 29): innovation consistency and tracking optimality point in opposite directions, so the frozen NIS-selected run closes at 0.638 at impact on the in-distribution test set and the consistency route to a larger band is not supported by the held-out training encounters. The mechanism is model error, not miscalibration: inflating the state-dependent noise compensates the forecast median's contraction, buying gain at the price of consistency, and the test-selected variant excluded by the freeze rule is disclosed, with its number, only in appendix C. The member-noise band of the production filter is 0.764 over five seeds. The legitimate gain comes from structure instead of noise: the two-stage filter of §3.3.3, at the production coverage band (a single validation-calibrated noise scale, no test-set tuning), reaches 0.794 at impact on the in-distribution test set (root-mean-square error 0.286) and carries the boundary test set as above, the best filter in the ladder among those selected without reference to the test set.

When a quarter-convective-time latency is acceptable, the smoother changes the deployment rule. The fixed-lag recursion of §3.3.4 on the linear stack cuts the impact error to 0.286 and the relaxation to 0.149 (filter values 0.375 and 0.141), a full-interval reanalysis at lag five; the ensemble smoother on the forecast-noise filter degrades it (supplementary material). Online estimation therefore takes the nonlinear filter and a $0.25\,t/c$ latency budget takes the closed-form linear smoother.

What the training objective confers survives to the assimilation stage, and per-family end-to-end evaluation makes the comparison fair: each family senses at its own taps, encodes its own observations, forecasts with its own operator and decodes through its own decoder. The wake-supervised predictive state halves the reconstructive anchor's assimilated load error in every phase (impact 0.26 against 0.41, relaxation 0.14 against 0.37; the lift-focused predictive state between at 0.31), while decoded-field fidelity is family-insensitive (figure 28a). The sensor axis separates the families further: the predictive state exploits every added tap monotonically where the reconstructive one saturates at $K = 8$, and it degrades more slowly with tap noise, still beating the clean reconstructive stack at twenty per cent noise (figure 28b,c). Streaming multi-encounter assimilation under a deployment-realism protocol, and a filter on the four-dimensional lift-critical state of §4.2.2, are reported in appendix B.

The closing study runs the whole design axis through one identical per-family end-to-end protocol: three families at four dimensions (figure 13). The predictive family is uniform, impact error 0.298 to 0.265 across $d \in \{4, 8, 16, 32\}$ with peak errors of ten to fifteen per cent everywhere; even its $d = 4$ cell (0.298) beats the linear floor's best (0.346 at $d = 32$). The linear basis is stable and a step worse everywhere. The reconstructive baseline is non-monotonic in dimension and irreproducible across seeds: it fails outright at $d = 4$ and $d = 32$ (1.620 and 2.247, peak errors above one hundred per cent, assimilation at $d = 32$ making the open loop worse), and its $d = 16$ cell, the best single cell in the grid at 0.180, does not survive seed replication: across three encoder seeds the same cell spans 2.252 impact error on average with individual seeds at 0.18, 0.65 and 5.9, the worst of them failing exactly as the neighbouring dimensions do. The mechanism is verified and lives on the observation side: linear probes on the true reconstructive latents are healthy at every dimension and every seed, while the pressure-to-latent map fails to recover the lift-carrying directions, insensitive to its regularisation, under a protocol that succeeds for the other two families in every cell. Wall-pressure estimatability of the reconstructive

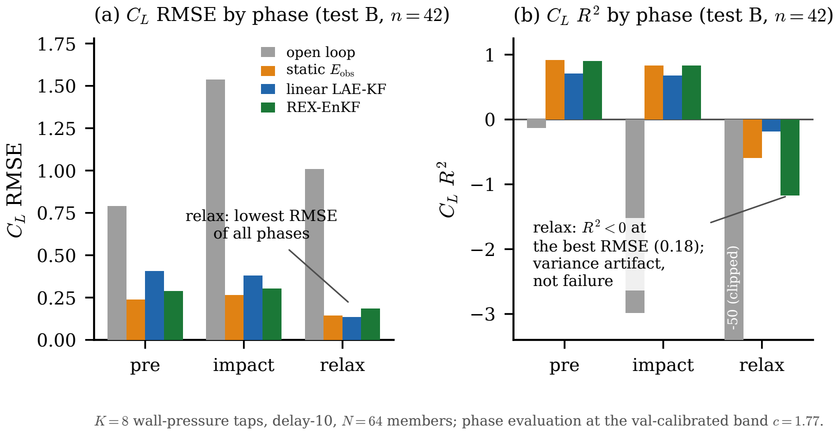


Figure 12. Phase-resolved lift tracking with the two-stage production filter (the estimator adopted for the performance envelope) on the in-distribution test set ($n = 42$ encounters, $K = 8$ taps, delay 10, $N = 64$ members, every-frame pressure). Lift error per phase in physical units (a) and in R^2 (b) at the val-calibrated band: the relaxation R^2 collapse is a variance artifact, the physical error the honest phase metric. The estimator ladder, its noise calibration and the fixed-lag smoother are in appendix C (figure 29).

geometry is unpredictable across both dimension and training seed; the anchored families are uniformly estimatable, which is the deployment-robustness argument of this paper.

Two loose ends close the section. The under-dispersion of the base filter has two regimes, not one: at impact it was calibration, and the state-dependent noise removes the consistency failures entirely; in relaxation it is model error, the forecast median's contraction, which no noise model fixes and the smoother does. And the wake-enstrophy null of the frozen filter, no family's filter tracks E_w, now has its mechanism: the wall constrains the lift-carrying near-body directions, the ones the Chang decomposition of §3.1.1 weights, while the wake enstrophy lives in shed structure the taps reach only through the delay window, so the null is the sensing geometry, not the state; the delay-for-sensors trade of §4.5 bounds what any wall-limited estimator of this flow can see.

5. Discussion

5.1. *What a wall-estimable state must contain*

The comparison has a direction, read off the chain of tasks a wall-borne reduced state must perform. On the decoded field the three families are indistinguishable, the energy-optimal linear basis as sharp as the nonlinear states, so field fidelity does not separate them. The separation opens at the first task that needs the wake, the observable read-out, where the linear basis loses the load at the impact instant and neither it nor the reconstruction reads the wake; and it widens at every step toward deployment. Through the impact the reconstruction's forecast collapses and the linear basis fades; recovering the state from sparse wall pressure, the predictive latent carries the most physics from a less recoverable state; and under assimilation at the strong-gust boundary the predictive filter is the only one whose innovations stay consistent on nearly every encounter. The predictive state holds furthest down the chain, and that is the design rule stated operationally: a state earns its place at deployment not by rendering the field but by remaining readable, forecastable, wall-recoverable and assimilable at once.

The central result is a design rule for a reduced state that a wall-pressure estimator can track. Such a state must carry the wake observables that govern the post-impact transient,

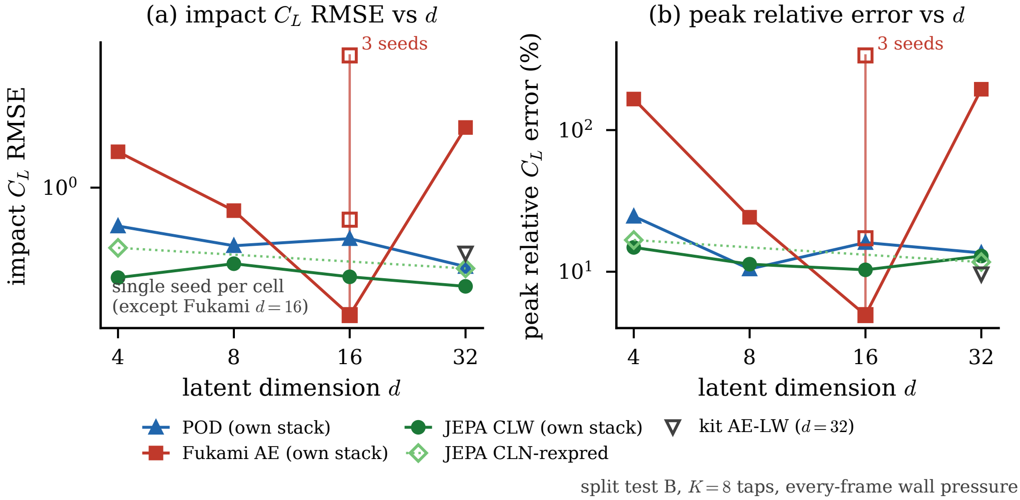


Figure 13. The assimilation-versus-dimension grid (in-distribution test set, per-family end-to-end stacks, $K = 8$, every-frame pressure; single seed per cell except the reconstructive $d = 16$ cell, which carries three): impact error and relative peak error against dimension for the three families. The predictive family is uniform, the linear basis stable, the reconstructive baseline non-monotonic in dimension and irreproducible across seeds.

must be organised so that a forward model can advance it without leaving its own support, and must be recoverable from the surface pressure that is the only measurement a vehicle carries. These three requirements are supplied by different ingredients, and separating them is the paper's contribution. Observable supervision on the wake supplies the first, the instantaneous readability, and, less obviously, much of the second: it concentrates the encoded variance into a low-rank subspace aligned with the observables, and the forward rollout, trained and living in that subspace, stays in distribution, where an anti-collapse regulariser applied to a bare reconstruction produces a near-isotropic latent with no protected subspace and a rollout that leaks into the directions the training data never constrained (§4.4.1). The mechanism is geometric, and it refines our earlier reading, which credited the anti-collapse term: a reconstruction objective fixes the latent only up to a diffeomorphism (Kelshaw & Magri 2024), the anti-collapse term on its own drives the latent toward isotropy (Broustail *et al.* 2026), and it is the supervision that pins the variance onto the observable-aligned directions the rollout respects. The multi-step predictive objective supplies the rest of the second requirement, the long-horizon stability that a single-step objective and an objective-free supervised encoder both lack, and, across retrains, the reliability of the forecast itself (§5.4). The third requirement, wall recoverability, is where energy and information part company: the reconstruction latent is the most recoverable in raw variance but the least in physics, while the predictive latent carries the most wall-recoverable physics (§4.5), so the coefficient state a controller would want is not the one a reconstruction objective produces.

The wake, and not the forces, is where these requirements separate the families, and the coordinate-level reading of §4.2.3 explains why. The forces are carried redundantly in every family, so an energy or reconstruction objective retains the force signature of an encounter without retaining the distributed combination of coordinates that carries the vorticity redistribution producing it. The gust parameters alone predict the forces well and fail on the wake at the readout horizon, so a state can look accurate on the load while carrying none of the physics the load is built from, which is exactly the failure mode the estimation results expose. The linear basis, for its part, remains genuinely useful:

it is competitive on the body forces, amplitude-accurate in reconstruction, and it holds the shedding clock as well as the predictive latent (§4.3); the result is not that POD is obsolete but that the POD coefficients do not achieve comparable linear readability of the wake-enstrophy target at any dimension we tested.

One caveat is inherited from the compactness-versus-forecast comparison that motivates this study (Solera-Rico *et al.* 2025): a more expressive predictor might narrow the forecast gaps between the families. Three controls argue that the ordering is a property of the representations rather than of an underpowered predictor. The shared forecaster was capacity-selected on the validation set, a sequence-LSTM variant included; the matched autoregressive transformer diverges on the published-recipe latents at equal recipe and budget, so its failure is not a capacity artefact; and a per-family tuned recovery estimator raises every family without changing their order (supplementary material). The orderings reported here are therefore floors set by the representation, not ceilings set by the predictor.

5.2. *The estimation demonstration and its limits*

The filter turns the design rule into a working estimator: from eight wall-pressure taps and no field measurement it tracks the load through the encounter and extends the usable gust-intensity range beyond every static inverse, realising the online-estimation pathway the extreme-aerodynamic control literature has named (Fukami *et al.* 2024; Mousavi & Eldredge 2025; Eldredge & Mousavi 2025). We are deliberate about its limits, and the first is one the family comparison forces: the in-range load-tracking is a property of sequential estimation on a nonlinear latent generally, not of this state, so the paper does not rest the representational claim on it. What the predictive state contributes to the estimator is the strong-gust boundary, the calibration margin, and a tracked state whose coordinates carry the wake; what no state yet contributes is a filtered wake estimate, since the analysis wake closure is below the mean predictor for every family. Its recoverability advantage over a raw high-dimensional field latent is real but modest (§4.5), so the case for the coefficient state rests on the physics-recovery ordering and the envelope, not on a large recoverability margin. Its point tracking is good but its uncertainty is under-dispersed at the strongest and widest gusts, where the innovations are temporally correlated and the ensemble collapses. The visibility analysis suggests the mechanism: the forward model has large gain along the data-unconstrained near-null directions it never learned to damp, and those inject spread-consuming noise during assimilation. This points to a calibrated process and observation noise model, rather than covariance inflation, as the outstanding requirement, and we report the filter as a feasibility demonstration with a characterised soft spot, not as a deployed estimator.

The delay-embedding view names the deployment knob and its limit. Because the encounter is low-dimensional, the state can be tracked from few taps read over a short window, which is the sensing budget a small vehicle could carry; the trade between tap count and window length is explicit for the static reconstruction (§4.5.1), so a sensor-poor platform can in principle compensate with a longer window within the encounter's own timescale. The frozen filter does not inherit that trade, because its innovation rank, not the embedding, is what binds at reduced tap count, and the two statements together sharpen the same conclusion: the noise model, not the sensor count, is the binding constraint. The limit that no budget escapes is the growth of the encounter's effective dimension with gust strength. Read through delay coordinates the encounter is an intermittently forced low-dimensional system (Brunton *et al.* 2017), the baseline shedding carrying the delay-coordinate dynamics and the vortex impact acting as a forcing whose amplitude is the gust ratio, which is consistent with the envelope being organised by $|G|$: no fixed sensing budget embeds the strongest, most three-dimensional encounters, which is why the envelope closes where it does (§4.6.2).

5.3. *Decodability versus estimability*

A reduced state can be optimised for two deployments that pull apart. A spatially resolved latent decodes the vorticity field more sharply than the coefficient state, but it is of far higher dimension than the wall measurement and so is not recoverable from a sparse set of pressure taps. The coefficient state decodes the field less sharply and is the object a wall-pressure filter can track. We therefore separate the two roles: the coefficient state is the estimable, forward-usable object this paper is about, and the field-resolved latent is a decodability instrument rather than an estimation target. Pooling the encoder onto the coefficient state costs field fidelity across every family, by -0.093 in decode similarity and -0.095 in fitted-operator forecast merit for the predictive latent (supplementary material), and it gains nothing in wall recoverability that the coefficient state does not already have. The state a controller would carry is therefore not the one a field-reconstruction objective produces.

5.4. *Limitations and outlook*

Five limitations bound the claims. First, the attribution of the instantaneous results is unglamorous: the wake readability is supplied by the observable supervision and its tie between the predictive and reconstructive wake-headed states is seed-robust (0.727 ± 0.021 against 0.760 ± 0.019). The shared-operator forecast merit of §4.1.2 shows the same picture dynamically: over three operator seeds the three wake-headed states are statistically indistinguishable, with overlapping case-clustered intervals. An earlier seed study of the fitted merit, run under the residual U-Net operator as a stability check, cautions the same way: the supervised control is seed-fragile (spread 0.163, with one healthy retrain in three collapsing), the predictive latent is tighter but not the sharpest (± 0.098), and the family means overlap. No fitted-operator margin between the wake-headed states is therefore decisive on its own. The defensible dynamical claim rests instead on the co-trained model and its rollout: the wake-supervised predictive state's own predictor forecasts better than the shared operator extracts from any latent at short horizon (§4.1.2), and the multi-step objective confers long-horizon stability and lower drift, not a larger instantaneous readability margin. Second, the wall-normal offset Y remains the least resolved release parameter because the training mass concentrates near $Y \approx 0$; the additional cases make it weakly readable where it was unreadable before (§4.3), and we note this without leaning on it. Third, the $|G| = 4$ extrapolation degrades for a physical reason: the encoder observes a single mid-plane vorticity slice that captures a substantially smaller fraction of the three-dimensional content there (§2.1), so the boundary test set marks where that representation becomes incomplete, not a parametric-generalisation claim, and the filter meets the same limit from the sensing side (§4.6.2). Fourth, the visualisation decoder is a diagnostic on the frozen encoder, never part of the predictive loss, and the pooled coefficient state trades decode sharpness for estimability by construction (§5.3). Fifth, and foremost for what comes next, the filter's calibration is the priority open problem: its uncertainty is under-dispersed at the strongest gusts, and the natural next step is a learned or physically motivated process and observation covariance, ahead of any closed-loop use.

These limitations come into sharpest focus against the use the architecture is meant for. The predictor is, by construction, a latent dynamics function of the kind an estimator or controller requires, the latent is orders of magnitude cheaper to advance than the field, and the filter demonstrates the estimation half of that loop from the wall alone. Closed-loop actuation is outside the present evidence: the predictor is trained to forecast observed encounters within the training envelope, and predicting how an encounter would differ under intervention is a task its training does not address. The contribution is narrow by design, a controlled comparison and a frozen-filter characterisation on a single parametric flow, and what it establishes is bounded accordingly: a wall-estimable, forward-usable coefficient state, the ingredients that produce it, and the gust-intensity envelope over which a fixed sensing budget can follow it. The natural next steps are the noise model above and the same matched-estimator protocol on other parametric flows, to test how general the estimability ordering reported here actually is.

A further direction concerns the wake target itself. The present descriptor fixes where the wake vortices sit and at which scales their energy resides, but its coordinates are generic. Weak-form dominant-balance analysis (Ahnert *et al.* 2026) suggests a physically grounded alternative: inner products of the conservation-form vorticity-transport terms against compactly supported test functions, so that each supervised coordinate is the local magnitude of a governing mechanism, evaluated without differentiating the data. Such a target is capacity-matched to the present descriptor by construction and would align the protected subspace of §4.4.1 with the mechanisms themselves; the controlled substitution is left to future work.

6. Concluding remarks

We asked whether a reduced state of an extreme vortex-gust encounter can be constructed to carry the wake dynamics, advanced in time, and recovered from wall pressure alone. On construction, under the predictive objective a scalar lift target is necessary for a non-collapsed latent representation and a wake target is what makes the state carry the wake observables; the near-body refinement that further sharpens the peak load is reported in §4.1. The collapse result concerns latent health and the peak-load result concerns a paired increment over an already lift-supervised baseline; the two rest on different comparisons.

On compression, the statement is two-tier: load information is approximately unchanged over the tested dimensions under a nonlinear probe and is linearly accessible at four coefficients, whereas moderate linear readability of the wake diagnostics appears from eight coefficients and stable reconstruction of the spatial wake structure requires approximately sixteen or more over the tested dimensions. On forecasting, one shared direct multi-horizon forecaster makes the wake-supervised states the most forecastable and the predictive states the most forecastable on the lift; a one-shot forecast remains usable over forty steps where autoregressive rollout degrades through error accumulation, and supplying the true gust parameters did not improve the direct forecaster for this dataset and model class, so the deployable route to the encounter is a model corrected by pressure rather than the release parameters.

On estimation, a two-stage ensemble filter sensing eight wall-pressure taps tracks the lift through the encounter, phase-resolved and in physical units, with impact-phase $R^2 = 0.794$ on the in-distribution test set and $R^2 = 0.837$ at the $|G| = 4$ boundary. Across latent dimension and training seed the predictive states remain uniformly estimable from the wall, whereas the reconstructive geometry does not, for a verified observation-side reason; this robustness, rather than any single configuration, is the deployment argument.

The design principle is that a nonlinear latent can be made forecastable and wall-estimable by training it predictively under observable anchoring, at no cost in compactness. Three results bound the claims. A static delay-embedded inverse matches the filter's in-distribution impact median, and the filter's advantage lies in the tails, at the boundary set and in continuous state tracking. The wake enstrophy is not reliably recovered by the tap counts, delay windows and estimators considered here, although the encoded state retains it, which points to the present observation configuration rather than to the state. And the mid-plane representation omits a substantially larger fraction of the three-dimensional vorticity content at the strongest gusts (four cases), where the filter's uncertainty consistency degrades even as the median tracking holds. The natural next steps are a calibrated process and observation noise model for the filter, and the same matched-estimator protocol on other parametric flows, to test how general the estimability ordering reported here is.

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Declaration of interests. The authors report no conflict of interest.

Data availability statement. The reduced-order-model and evaluation code, the configuration files, the v2.2 data-split manifest, and the analysis outputs that regenerate the tables and figures are deposited at `https://doi.org/10.xxxx/zenodo.PLACEHOLDER` under the MIT License (DOI to be finalised on acceptance). The processed per-encounter cache, or a representative subset sufficient to rerun the heavier steps, is available from the corresponding author. The raw direct numerical simulations were computed with the SOD2D solver (Gasparino *et al.* 2024) and are owned by the simulation collaborators.

Author contributions. *[Authors to insert contributions, for example in CRediT format.]*

Use of generative-AI tools. Generative-AI assistance (Anthropic Claude, accessed through the Claude Code command-line interface) was used during 2026 to help with data analysis, figure generation, and manuscript drafting. All numerical results were computed by the authors' own code and verified against the underlying data, and the authors take full responsibility for the content of the article.

Appendix A. Architecture, anti-collapse regularisation, and uncertainty quantification

Architecture detail

The encoder is a hybrid convolutional stem followed by a small vision transformer. Three downsampling convolutional stages map the single-channel 192×96 vorticity field to a 24×12 feature map at 256 channels, that is 288 image tokens, which a six-layer pre-norm transformer of width 256 with eight heads processes; the latent is read from a prepended class token through a one-layer projection whose output passes a BatchNorm, the layer the characteristic-function regulariser below requires. The encoder is unconditional and carries no information about the gust parameters. The training predictor advances the pooled 32-vector directly: its function class is a six-layer autoregressive transformer of width 384 with sixteen heads, rotary temporal positions and a causal mask, mapping a short history of 32-vectors to the next 32-vector, with no spatial map or tiling anywhere in the loop. Training minimises an open-loop rollout loss over eight steps, with no teacher forcing, against online encoder targets that are detached (no exponential-moving-average target network), plus the regulariser below, with AdamW and bf16 mixed precision. The predictor advances the latent from its own history alone, with no gust input, so the gust parameters $\mathbf{c} = (G, D, Y)$ enter nowhere in the model. Two downstream forecast operators are fitted afterwards on each frozen latent, identically across families. Every family produces a pooled coefficient vector at the same dimension, and two matched forecast operators apply to all of them: an autoregressive transformer on the vector sequence, and a residual U-Net that tiles the vector to a grid and forecasts it convolutionally. The forecast merit of table 2 reports each latent under the operator that is numerically stable on it. The transformer is stable across the pooled states of the controlled matrix, forecasting the predictive, supervised, and reconstructive-autoencoder latents comparably and placing the wake-supervised predictive state at the head of that block; but refitted with the same recipe and budget on the reference latents (the β-VAE and the two Fukami autoencoders, which are plain convolutional autoencoders without skip connections) it diverges to large negative merit. The residual U-Net is stable on every latent, so it is the operator reported for the reference latents; imposing either operator on all families would either penalise the pooled states off-class or break on the references. The ensemble filter of §3.3 uses the same matched transformer on the wake-supervised predictive state's own pooled trajectories, where it is stable, and the wake-supervised predictive state's own co-trained predictor, the co-trained reduced-order model, is reported alongside. The narrow spread of the pooled block under the transformer, and the substantial seed variation of the merit, are why we read the fitted-merit ordering as a level with a small predictive edge rather than a decisive separation.

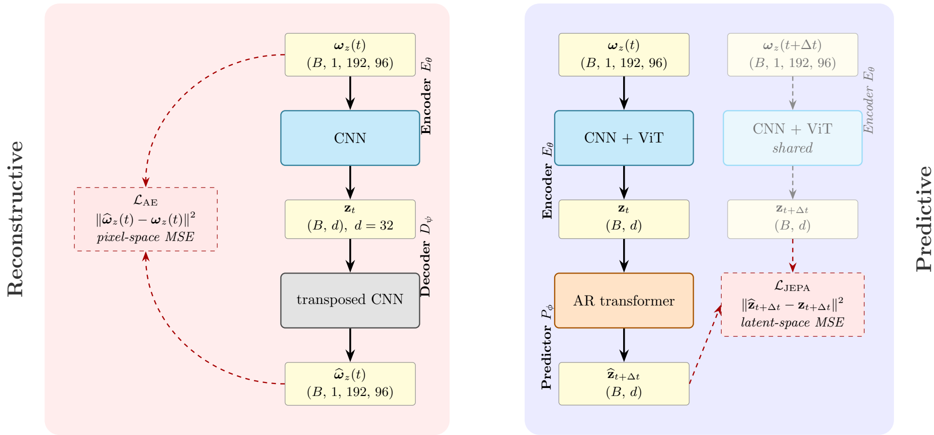


Figure 14. Predictive versus reconstructive training. Left: the reconstructive autoencoder encodes the field at t, decodes it, and incurs a field-space error. Right: the predictive model encodes the fields at t and $t + \Delta t$ with shared weights and incurs a latent-space error between the predicted and observed target embeddings, with no field-space loss and no decoder during training. This is the contrast the matched-predictor protocol of §3.2 evaluates; the visualisation decoder is trained only afterwards, on the frozen encoder.

Anti-collapse regularisation

Because the encoder and predictor are trained end to end on latent-space prediction with no decoder, the latent is held away from a collapsed solution by an explicit regulariser rather than by a reconstruction term. We use SIGReg, the characteristic-function (Epps–Pulley) regulariser of the from-pixels joint-embedding line (Maes *et al.* 2026; Balestriero & LeCun 2025), with M random one-dimensional projections of the latent and a fixed weight $\lambda_S = 0.02$ in the loss, the value used in every run reported here. The same regulariser has since been adopted independently for biosignal time series, where it is likewise reported to drive the embedding toward isotropy so that a structuring objective is needed alongside it (Broustail *et al.* 2026), consistent with the supervision-anisotropy mechanism of §4.4.1. As a safeguard the participation ratio of the latent is monitored every thousand iterations, with an automatic fall-back to a variance-covariance objective (Bardes *et al.* 2022) if it indicates collapse while the parameter probe is still uninformative; on the runs reported here the participation ratio stays healthy and the fall-back does not trigger.

Uncertainty quantification

Every held-out coefficient of determination and mean absolute error in the paper carries three independent uncertainty signals, following the reporting protocol fixed with the split. The first is a bootstrap confidence interval from 2000 resamples over the held-out test encounters, which is the dominant source of scatter at the present sample size ($n = 42$ for the in-distribution test set, $n = 40$ for the boundary test set) and is wide for the individual wake observables, as noted in §4.1.2. The second is the variance across three encoder retrains from independent random seeds, reported as the standard deviation in the objective-times-architecture controls of §4.4.1. The third is a five-fold cross-validation over cases on the read-out (probe) step, which guards against a probe that overfits the particular held-out split. Together these approximate what a full five-fold case-level cross-validation of the entire pipeline would buy, at a fraction of the training cost; a single split with bootstrap intervals is the community standard for this class of study.

Training-set fit

The predictive latent attains the highest in-distribution training fit of the forecast rollout among the families, but this is an in-distribution number, not the held-out result: the family ordering is established by the held-out representational closure of table 2, and the held-out forecast mean is markedly lower than the training fit, as it must be. We therefore do not tabulate the training-set fit, which would invite reading an in-distribution number as a result.

Table 9. Paired per-encounter tests for the estimation endpoints, on the frozen filter records. The family was fixed before any statistic was computed: the primary family F1 compares the filter analysis against the static single-frame recovery and F2 against the field-free open-loop forecast, on the endpoints $\{C_L, E_w\}$ in the impact and relaxation windows, one-sided with the filter declared in advance as better; $\mathit{\Delta} R^2$ is the median per-case mean improvement, the p is a case-level Wilcoxon signed-rank, and Holm corrects within each family. On the pooled in-distribution test encounters the filter beats the open-loop forecast decisively on the load but not the static recovery, which is accurate at the moderate gust strengths the in-distribution test set samples; the filter's advantage over the static delay-embedded inverse concentrates at strong gusts, where the boundary test annex (characterisation only, $n = 8$ cases) is decisive on the load in both windows. The wake enstrophy through the filter does not beat the static recovery at the boundary, a null result.

Family	Endpoint	Window	median $\mathit{\Delta} R^2$	Wilcoxon p	Holm p
F1 (vs static)	C_L	impact	0.16	0.161	0.483
F1 (vs static)	C_L	relax.	0.53	0.116	0.465
F1 (vs static)	E_w	impact	−25.15	0.986	1.000
F1 (vs static)	E_w	relax.	−2.47	0.839	1.000
F2 (vs open loop)	C_L	impact	9.43	0.001	**0.003**
F2 (vs open loop)	C_L	relax.	44.84	0.001	**0.003**
F2 (vs open loop)	E_w	impact	17.39	0.246	0.246
boundary test annex	C_L	impact	1.11	0.0078	–
boundary test annex	C_L	relax.	6.41	0.0039	–

Paired tests on the estimation endpoints

Table 9 reports the per-encounter paired family behind the tracking and envelope claims of §4.6.1 and §4.6.2. Pairing cancels the encounter-to-encounter difficulty that widens marginal intervals, and clustering by case respects the shared shedding history of a case's encounters.

Topology of the encoded encounter

A coordinate-free check on the geometric mechanism of §4.4.1: Vietoris–Rips persistence of the encoded trajectories, per family, with a generator declared significant when its persistence exceeds a noise floor set at a fraction of the cloud diameter, after per-family standardisation so that coordinate scale cannot carry the result. On the gusted held-out encounters the supervised states keep a single clean H_1 cycle in about half the encounters (0.64 for the predictive and 0.52 for the objective-free supervised state) where the anti-collapse reconstruction fragments (0.10); on the single-period no-gust control the same separation holds (0.62 against 0.00), and it survives whitening, so the fragmentation is a property of the reconstructive encoding rather than of any coordinate scale. The linear basis keeps the cleanest no-gust loop of all (0.62), consistent with a linear projection of a smooth field; the claim is that the anti-collapse reconstruction fragments, not that only the predictive latent carries a loop.

Preprocessing robustness

The wake enstrophy squares the vorticity, so the per-encounter high-percentile amplitude clip of the ω_z preprocessing could in principle shape the primary endpoint. We test this on the frozen encoders, with no retraining, by recomputing the readout-frame wake-enstrophy target under three training-only clip treatments of the raw field and re-fitting the frozen-latent-to-wake ridge probe per family: no amplitude clip, the production per-encounter $p_{99.99}$ clip, and a single leak-free $p_{99.99}$ threshold pooled over the training encounters. The predictive latent leads the wake on the in-distribution test set under all three (table 10), with the largest shift in the predictive-minus-best-baseline advantage relative to the production clip being 0.07 in R^2. The leak-free training-global clip gives the largest predictive advantage, so the per-encounter clip is not the source of the wake result.

A.1. *The near-body conditioning increment*

On top of the lift anchor the near-body force-element (Chang) head confers real peak-load accuracy, the strongest cell of the $2 \times 2 \times 2$ conditioning cube of §4.1: the lift-focused predictive state (cube code CLN) beats the lift-only cell CL by 2.15 peak-region R^2 points with a case-clustered interval that excludes zero ([0.03, 5.59]), at a pooled peak-region $R^2 = 0.86 \pm 0.003$ over three seeds, the tightest seed band anywhere in the cube, and

Table 10. Preprocessing robustness of the representational wake closure, established on the previous-generation spatial encoders (d = 64, the 85-case v2.1 split) and retained as evidence that the amplitude clip does not shape the endpoint; the pipeline is unchanged in the present dataset revision, and the probe protocol is identical. Held-out in-distribution test wake-enstrophy R^2 from a frozen-latent ridge probe (no retraining), recomputed under three training-only amplitude-clip treatments of the raw ω_z field. The predictive latent leads the wake under every treatment; the advantage keeps its sign and approximate magnitude.

Family	no clip	per-encounter	training-global
predictive	0.75	0.80	0.84
reconstructive (Fukami)	0.12	0.10	0.08
linear (POD)	-0.27	-0.05	0.00

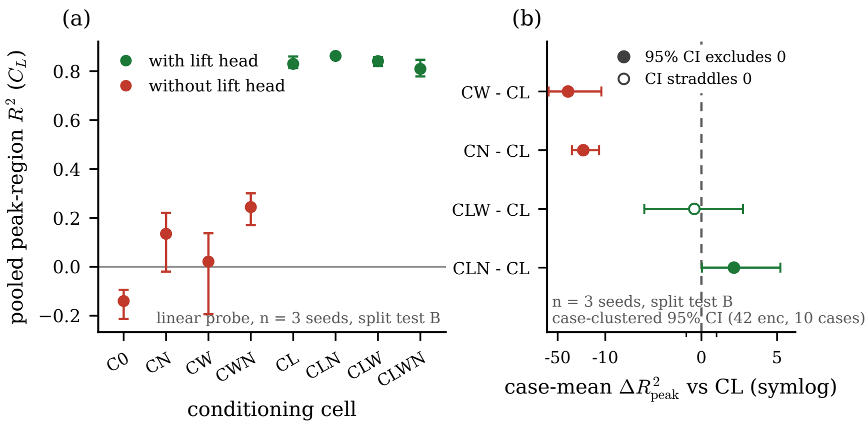


Figure 15. Paired peak-region lift R^2 per conditioning-cube cell (left) and case-mean deltas against the lift-only cell CL (right, symmetric-log axis, 95 % case-clustered bootstrap intervals over the 42 in-distribution test encounters of 10 cases); filled markers mark intervals that exclude zero. The near-body lift-focused predictive state cell (CLN) is the unique positive increment; the no-L cells sit one to two orders of magnitude lower.

a decoded phase lag of -0.011 t/c (figure 15). The head construction and its pre-committed quality gate are described in §3.1.1. Downstream the wake-headed wake-supervised predictive state is the single predictive representative; the near-body head is a lift-sharpening refinement, not the primary state.

A.2. *Field-decode and phase-resolved figures*

A.3. *Methods and comparison tables*

Appendix B. Sparse wall-pressure state estimation

At deployment the vorticity field is not available; only the wall pressure is. This appendix records the sensing conventions behind §4.5 and §4.6.1 and the placement-robustness checks that the main text summarises. The input throughout is the span-averaged pressure on the 192 surface taps; the recovery estimator is selected per family by encounter-grouped cross-validation among a kernel-ridge regressor on the flattened window, a multilayer perceptron, and a recurrent network over the window sequence, and the recovered state is read by the same frozen probes as everywhere else.

Placement, and why it does not drive the ordering

Two placements are used, deliberately. The filter and the recovery of table 5 sense at each family's own model-conditioned taps, the greedy selection that best exposes that family's latent, because the estimator comparison should give every family its best sensing. The delay-trade grid of §4.5.1 instead uses a single target-blind array, the QR-pivot optimum of the wall-pressure modes (Manohar *et al.* 2018), nested so that every tap count is a prefix of the next, because a sensor-budget claim must not depend on a placement tuned to any model. The two meet where their sensing coincides: the grid's eight-tap, thirty-frame cell rerun on the model-conditioned taps

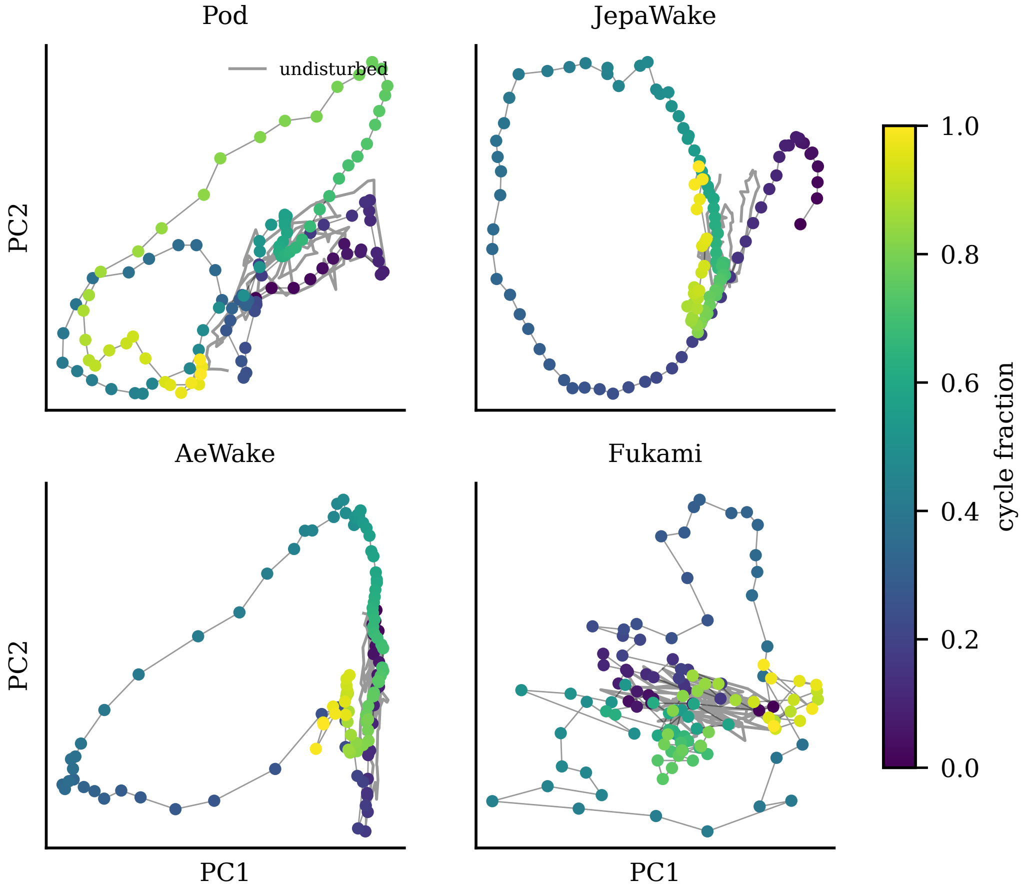


Figure 16. Latent trajectory portraits by family (leading two principal components of the $d = 32$ state; a representative gusted encounter coloured by cycle fraction, undisturbed limit cycle in grey). The predictive, linear and matched-supervision states trace coherent cycles perturbed by the gust; the published-recipe reconstruction does not.

Table 11. Encoder and training-predictor configuration, counted from the released $d = 32$ checkpoint. The model is unconditional: no gust parameters $\mathbf{c} = (G, D, Y)$ enter the encoder or the predictor. Two small auxiliary heads read observables off the encoder output during training (a lift head and an 80-dimensional wake-spectrum head; §3.1). The estimation-stage forecast model of §3.3 is a predictor of the same class refitted on the frozen pooled trajectories.

	Encoder E_θ	Predictor P_ϕ
role	field $\longrightarrow$ pooled state (unconditional)	pooled-state rollout (unconditional)
input	$\omega_z \in \mathbb{R}^{192\times 96}$	pooled states $\mathbf{z} \in \mathbb{R}^{32}$
backbone	3 conv. stages + 6-layer ViT	6-layer causal transformer (RoPE)
width / heads	256 / 8	384 / 16
image tokens / context	$24 \times 12 = 288$	2 frames, rolled 8 steps open loop
conditioning	none	none
state read-out	class token $\longrightarrow$ linear $\longrightarrow$ BatchNorm	next pooled state
parameters	6.7M	10.7M

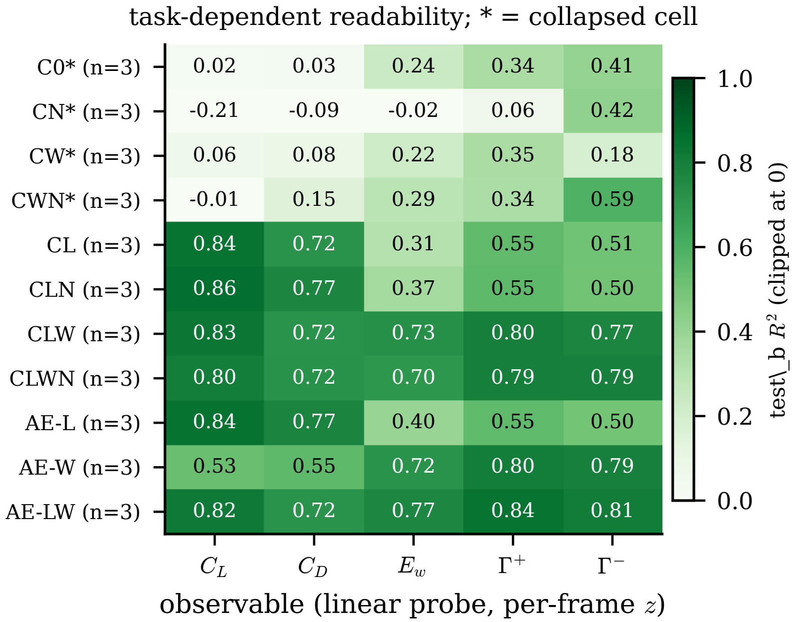


Figure 17. Task-dependent readability (in-distribution test set, linear probes on the per-frame state, seed mean over three encoder seeds per cell): lift and drag follow the L head, the wake block (E_w, $\Gamma^{\pm}$) follows the W head or the reconstruction objective, and collapsed cells (asterisks) read nothing linearly.

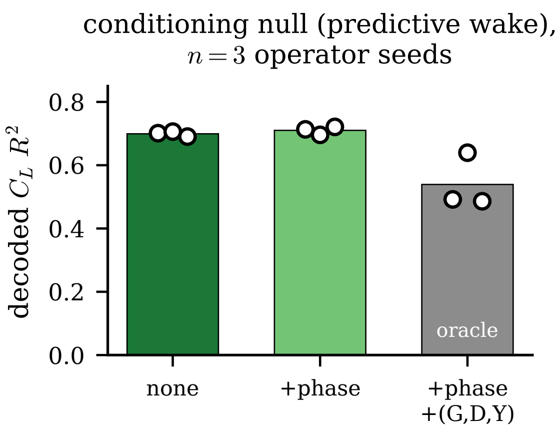


Figure 18. The conditioning null on the shared direct multi-horizon forecaster (demoted from the forecast figure 7): decoded lift R^2 of the tuned forecaster on the wake-supervised predictive state latents under no covariate, an added phase covariate, and an added (G, D, Y) gust-parameter oracle (three operator seeds per arm). The deployable phase covariate is a wash with the unconditioned forecaster, while the gust-parameter oracle degrades it.

reproduces the table's recovery exactly (0.656). The energy-versus-information split of §4.5 is also placement-robust: under the shared target-blind array with the kernel-ridge estimator alone, the reconstructive state remains the most recoverable in variance (0.775 against 0.667 for the predictive and 0.500 for the linear basis) and the predictive state retains the most recoverable physics (0.523 against 0.363 and 0.391), so neither ordering is an artefact of the model-conditioned placement.

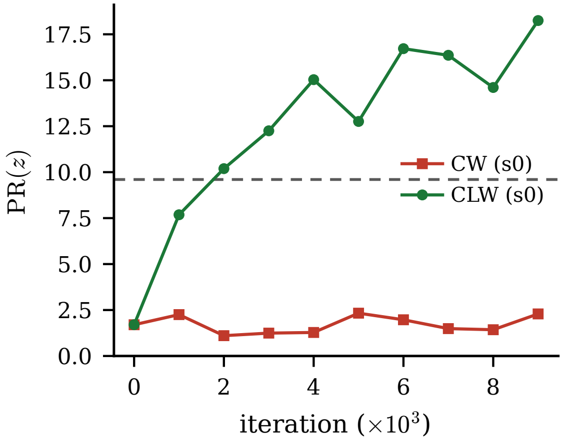


Figure 19. Training history of the conditioning-cube collapse (demoted from figure 4): participation ratio against iteration for one representative collapsed cell (CW, seed 0) and one healthy cell (CLW, seed 0), with the collapse flat from the first diagnostic while the healthy cell crosses the floor within the first quarter of the run.

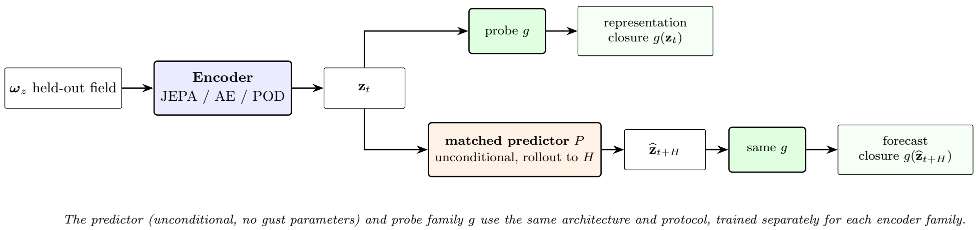


Figure 20. The evaluation protocol, applied identically to every encoder family (demoted from the framework figure 3). The same matched forecast operator is refit and the same probe family g is applied to every encoder: a probe on the simulation-encoded latent measures representational closure, the headline, and the same probe on the rolled-out latent $\widehat{\mathbf{z}}_{t+H}$ measures forecast closure, reported as secondary.

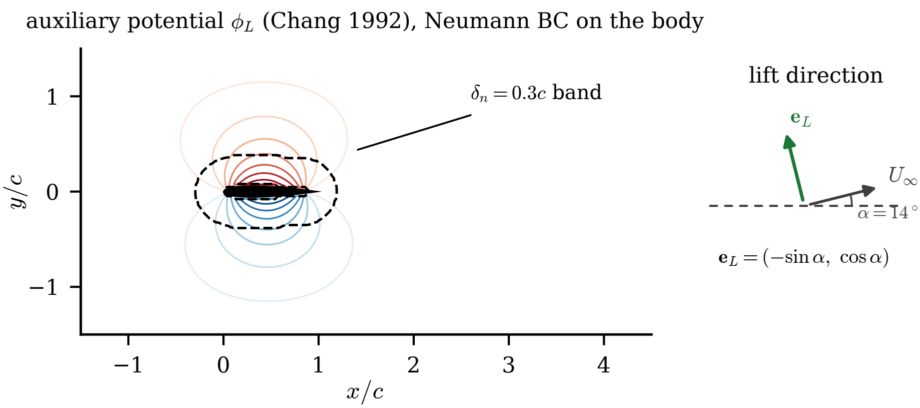
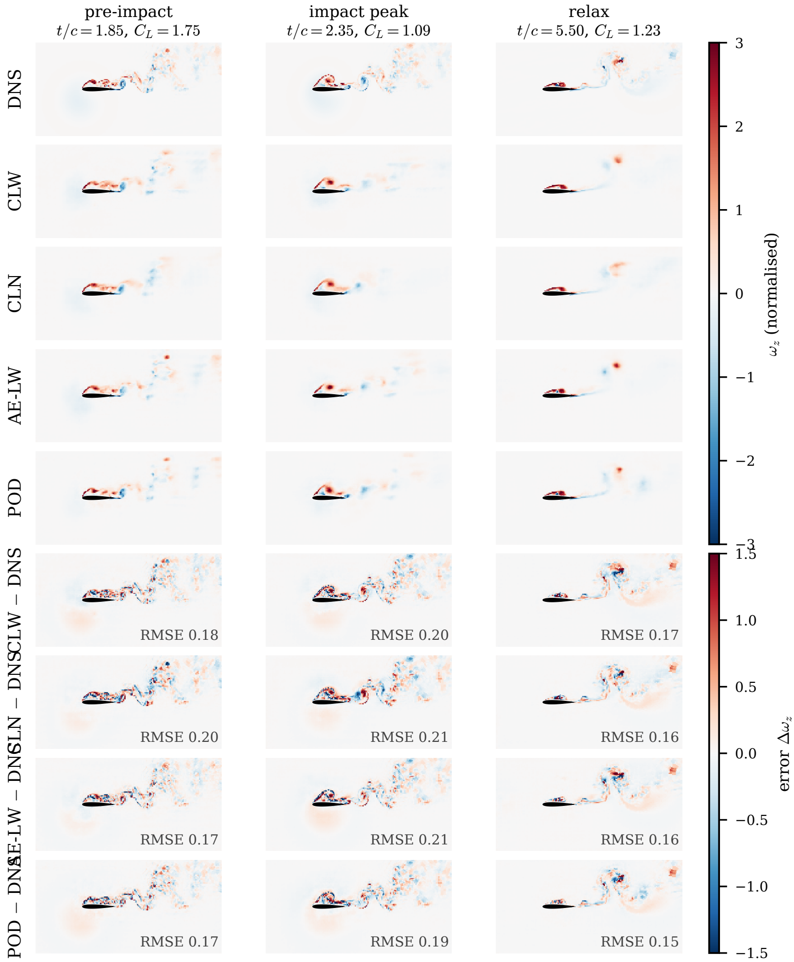


Figure 21. The Chang auxiliary potential ϕ_L of equation (3.1) on the cache grid (contours), the feathered $\delta_n = 0.3c$ supervision band (dashed) and the airfoil footprint; the inset shows the lift direction $\mathbf{e}_L$ against the inclined freestream at $\alpha = 14°$.



case G-0.50_D1.00_Y-0.40 (test_b), encounter 0; seed s0 per model; pooled z tiled to the 24 x 12 grid, frozen-encoder decode-floor decoders

Figure 22. Decode floor on a representative held-out in-distribution test encounter (one decoder seed per model): DNS, the wake-supervised predictive state, the lift-focused predictive state, the wake autoencoder and the linear basis, and the error rows at three phases. Identical-capacity frozen decoders; normalised vorticity at the fixed ±3 range. All four states render the field faithfully, including the linear basis: field decode is not where the states separate.

What the window recovers that the instant cannot

At a single frame the wall is blind to the wake-circulation direction of the predictive state (§4.5); over the thirty-frame window the blind coordinate becomes partially recoverable (0.33 held-out, from near zero at one frame), and the wake enstrophy read through the recovered state reaches 0.40. The window, not the tap count, is what carries this: the same quantities barely move as taps are added at a fixed single frame. This is the delay-coordinate content of the sensing result, quantified in the grid of §4.5.1; it is a statement about the static reconstruction, and the filter inherits the sequential version of it through its assimilation window rather than through any explicit delay embedding.

The recovered field

Decoding the pressure-recovered state through each family's diagnostic decoder makes the recovery visible in physical space (figure 25). The comparison within each row, recovered field against that family's own decode ceiling, isolates the pressure step from the decoder's reconstruction limit: the loss from ceiling to recovery is

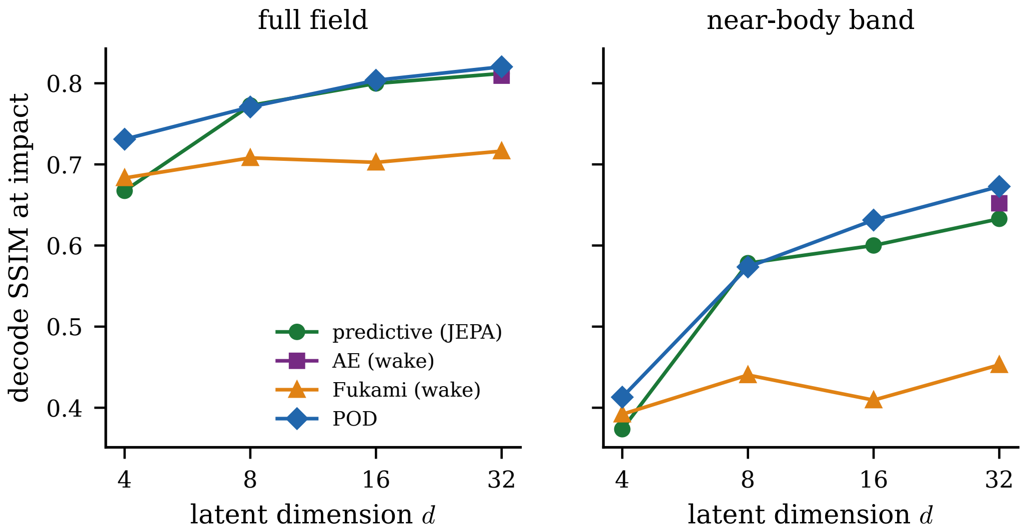


Figure 23. Decode SSIM at the impact instant against latent dimension d (full field and near-body band, in-distribution test set, decode floor). The compact nonlinear states and the linear basis improve with d and saturate together; only the published-recipe lineage plateaus lower. Field decode is not the discriminator at any dimension. Single encoder/decoder seed.

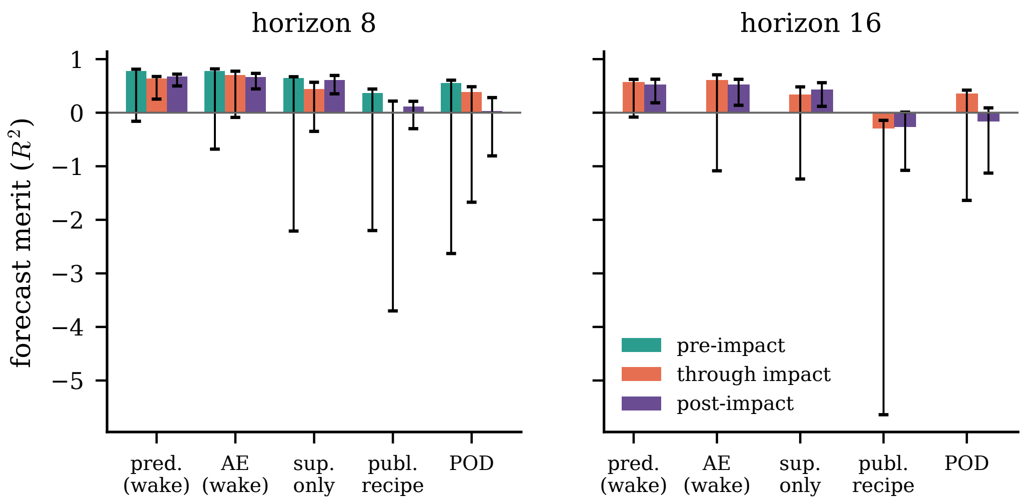


Figure 24. Shared-operator forecast merit split by phase (pre-impact, through-impact, post-impact), per family at horizons eight and sixteen (in-distribution test set, five observables, three operator seeds; bars at the seed mean, case-clustered 95 % intervals). Through-impact merit separates the families: the wake-supervised states hold, the published-recipe reconstruction collapses across the transient, and the linear basis fades into the relaxation.

the part attributable to the eight-tap window, and the panels show it concentrating in the wake structures rather than in the near-body field, consistent with the wall reading the force-carrying directions best (§4.5).

The out-of-distribution boundary from the sensing side

At the $|G| = 4$ boundary every static approach fails on the load, the via-latent recovery at −0.33 and the direct no-latent regression at 0.08 from the same taps (0.14 with the full array), while the filters still track it (table 6). The boundary is the same larger three-dimensional content the field carries there (§2.1): the mid-plane state

Table 12. The compared families: encoder, latent, and training objective. Every family produces a pooled $d = 32$ coefficient vector at the same dimension. The top block is the controlled matrix, one shared encoder (the hybrid convolutional-stem plus vision transformer of table 11) with the training ingredient varied one at a time; the objective-free supervised control trains that encoder with the observable heads and the anti-collapse term alone, with no predictor and no decoder, so it isolates what the supervision by itself supplies. The bottom block is the published reference recipes: the Fukami autoencoder and the β-VAE are plain convolutional autoencoders, a convolutional stack down to a fully-connected bottleneck with no skip connections, and POD is a linear basis. The forecast merit of table 2 reads each latent through a separate matched forecast operator; the operators (an autoregressive transformer and a residual U-Net) and the reason a different one is stable on the reference latents are described there and in Appendix A.

Family	Encoder	Latent	Training objective	Source
predictive	CNN + ViT (shared)	pooled 32-vec	latent-space prediction + heads + SIGReg	this work
supervised only	CNN + ViT (shared)	pooled 32-vec	heads + SIGReg only (no objective)	this work
reconstructive AE	CNN + ViT (shared)	pooled 32-vec	field reconstruction + heads + SIGReg	this work
anti-collapse AE	CNN + ViT (shared)	pooled 32-vec	field reconstruction + SIGReg (no heads)	this work
Fukami AE	conv. AE, no skips	pooled 32-vec	reconstruction + observable head	Fukami & Taira (2023, 2025)
β-VAE	conv. VAE, no skips	32-vec (mean)	reconstruction + KL + heads	Solera-Rico *et al.* (2024)
POD	linear projection	32 leading modes	none (SVD basis)	classical

Table 13. Configuration of the pressure-only ensemble Kalman filter in the latent space (§3.3, figure 3b). Every setting is frozen before the held-out evaluation. The multiplicative inflation is selected per family on the validation split, since the families differ in dispersion: the predictive and linear filters select no inflation ($\rho = 1.00$), being under-dispersed, while the reconstruction filter selects $\rho = 1.05$.

Component	Choice
Filter	stochastic ensemble Kalman filter, latent space
Ensemble size N	64 members
Forecast model	matched autoregressive transformer, refit on frozen train trajectories
Observation operator $\mathcal{H}$	$\mathbf{z} \mapsto K$ span-averaged wall-pressure taps
Sensor count K	8 taps, per-model optimal (OSP) placement
Process noise $\mathbf{Q}$	from the predictor's one-step residual covariance
Observation noise $\mathbf{R}$	calibrated from held-out pressure residuals
Inflation ρ	validation-selected per family (1.00 predictive/linear, 1.05 reconstruction)
Initialisation	field-free: windowed pressure-to-latent regressor + its residual covariance
Assimilation window	[impact − 24, impact + 48] frames
Leakage guard	lift and wake probes never enter the innovation (enforced, unit-tested)

Table 14. Where each rolled state goes relative to the training distribution, on the in-distribution test set at rollout horizon sixteen. The latent covariance condition number measures anisotropy, and the near-null departure fraction is the share of the rollout departure that falls into the bottom-quartile directions of the encoded covariance. An isotropic null would leak 0.25. The anti-collapse reconstruction is near-isotropic and leaks; the supervised states are anisotropic and stay in their low-rank subspace.

State	covariance condition number	near-null departure fraction
anti-collapse AE	11	0.100
predictive (wake)	451	0.014
supervised only	1041	0.029

captures less of the dynamics, so a static delay-embedded inverse recovers little from a single instant, and only the accumulated pressure history carries the load through.

Table 15. Per-estimator configuration (Gupta *et al.* 2026, table 1 analogue). All fits use the training split only; ρ and the two c calibrations use validation-family splits as stated in the text and nothing downstream of them. Ensemble size $N = 64$, taps $K = 8$ at the family's own placement, delay $m = 10$, every-frame assimilation, and the consistency-failure criterion of §3.3.2 are common to all rows.

estimator	dynamics F	observation h	process noise Q_t	noise scale
static delay-embedded inverse	none	E_{obs}, eq. (3.8)	none	none
base EnKF (frozen)	matched transformer	linear $\mathbf{z} \to \mathbf{p}_K$	teacher-forced residual cov.	$\rho = 1.00$ / 1.05
linear latent filter	linear A (SV-clipped)	E_{obs}, $h = \mathrm{Id}$	A-fit residual cov.	$\rho = 1.0$
forecast-noise filter	forecaster median, eq. (3.9)	E_{obs}, $h = \mathrm{Id}$	quantile band, eq. (3.9)	$c = 1.77$ (coverage)
two-stage filter	median forecast inside impact, transformer outside	E_{obs} inside, tap space outside	as per stage	c as the forecast-noise filter
fixed-lag RTS	linear A	E_{obs}, $h = \mathrm{Id}$	A-fit residual cov.	lag $L = 5$

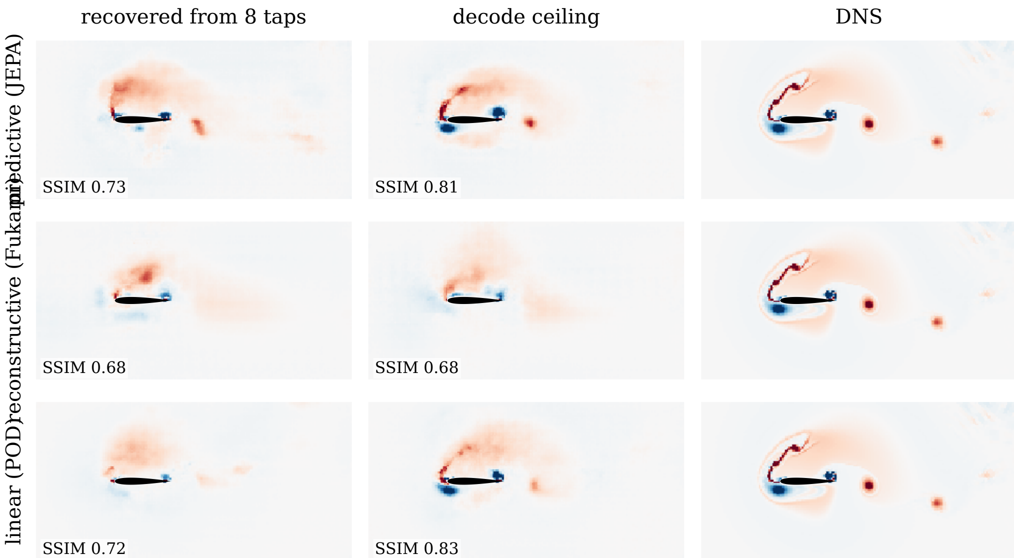


Figure 25. Flow recovered from eight wall-pressure taps on the representative held-out encounter of figure 10, at the impact-side frame, one row per family. Columns: the field decoded from the pressure-recovered coefficient state, the decode ceiling (the same decoder applied to the encoded state, no pressure step), and the simulation; per-panel structural similarity annotated. The comparison is within each row, since the decoders' ceilings differ (§5.3); the ceiling-to-recovered gap is the cost of the pressure step.

Delay-coordinate knobs

The window sweeps of §4.5.1 keep the delay stride at the cache cadence ($\Delta t = 0.05\, c/u_\infty$), and this choice is a convention rather than a tuning: the first minimum of the time-lagged mutual information of the tap signals (Fraser & Swinney 1986) sits near 29 frames, about one shedding period, so the thirty-frame window spans roughly one decorrelation time of the wall pressure at the cadence used. The window length itself is selected by the measured recovery trade rather than by a false-nearest-neighbours criterion (Kennel *et al.* 1992), which we regard as a cross-check and not a design input: the delay-embedding guarantees are generic, wall pressure is a physical and non-generic observable strongly coupled to the near-body vorticity, and the operative evidence is the measured recovery, with the theorem as the organising principle rather than a proof that a given tap count suffices.

Streaming deployment realism and the four-coefficient filter

Under a deployment-realism protocol, streaming multi-encounter assimilation with a pressure-only initialisation and no oracle anywhere, the filter on the wake-supervised predictive state holds an impact-phase lift closure of 0.789 on clean taps and 0.780, 0.749 and 0.646 at five, ten and twenty per cent tap noise (three member-noise seeds each, figure 26). The four-dimensional lift-critical state of §4.2.2 supports a filter of its own at 0.782 impact median over five seeds, so a four-coefficient state is estimable from the wall as well as forecastable.

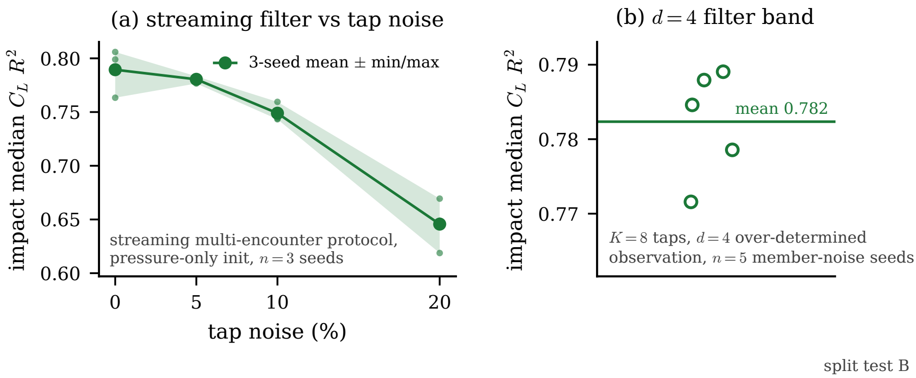


Figure 26. Deployment realism (in-distribution test set): (a) streaming multi-encounter filter against tap noise, three member-noise seeds, validation-selected band (no test-set reference); (b) the $d = 4$ filter band over five member-noise seeds.

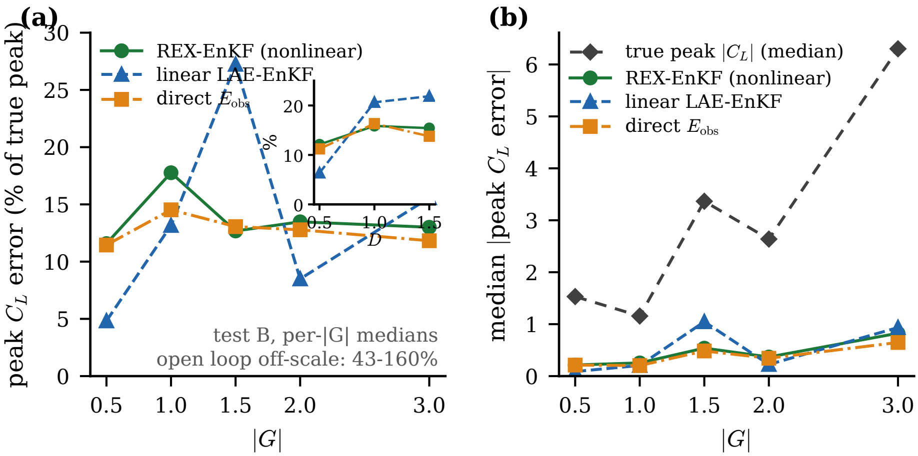


Figure 27. Relative peak error across the gust envelope (in-distribution test set, medians per $|G|$): the forecast-derived filter is scale-invariant near 12 to 14 per cent while the linear filter is non-uniform; the absolute peak error (right) grows with the peak itself.

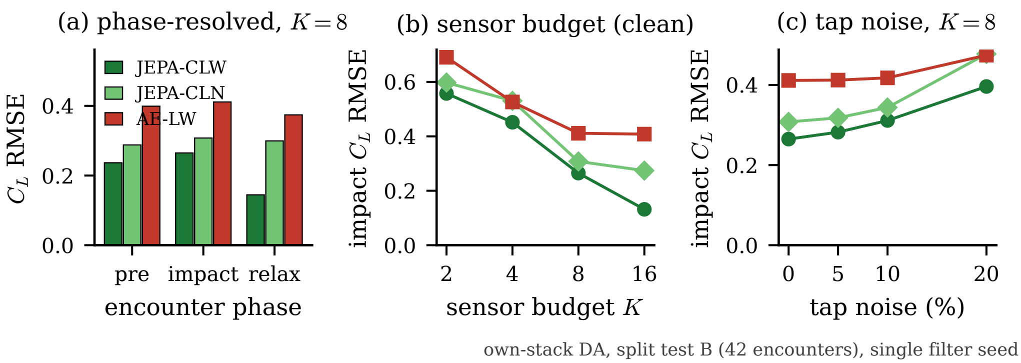


Figure 28. Per-family end-to-end comparison (in-distribution test set, 42 encounters, single filter seed): (a) phase-resolved lift error per family; (b) sensor-budget sweep on each family's own nested placement; (c) tap-noise sweep at $K = 8$.

Choosing among the estimators

The ladder of §4.6.3 supports explicit selection guidance, in the spirit of the method-choice discussions of the assimilation literature (Gupta *et al.* 2026). When only the in-distribution median matters and latency is irrelevant, the static delay-embedded inverse is the strongest and simplest tool, and no dynamics are needed. The moment the tails, the extrapolation boundary or continuous state tracking matter, the filters take over: online, the two-stage filter is the working choice, the forecast-derived noise carrying the impact and the tap-space observation carrying the relaxation; with a quarter-convective-time latency budget, the closed-form fixed-lag smoother on the linear stack is both the cheapest and the best reanalysis. Sparse or intermittent pressure favours the linear filter with encoded observations, the only member with zero consistency failures and graceful degradation under subsampled updates; every-frame pressure is necessary for the nonlinear members. Noise calibration should be spent on structure before scale: the state-dependent band fixed the impact-side consistency failures outright, while inflating its scale toward better tracking is model-error compensation that innovation consistency rejects, so a larger band should be read as a diagnostic of forecast contraction, not as a calibration target. Across model families the guidance is one sentence: anchored latent geometries, predictive or linear, can be selected on dimension and budget alone, while unanchored reconstructive geometries require a per-configuration estimatability check that their probes do not predict.

Appendix C. Calibration disclosure

The forecast-derived filter's band scale carries a full audit trail. The production value $c = 1.77$ is the coverage calibration of §3.2.1, selected on validation before any test encounter was seen. During exploration a band of $c = 4$ was evaluated against the in-distribution test set and reached 0.840 impact median there; because that value was selected with the test split in view, it is excluded from every table and headline of this paper and quoted only here. The consistency calibration (grid and rule committed before any value was seen) subsequently showed that the innovation statistic rejects large bands on held-out training encounters even though their tracking is better (§4.6.3): the large-band gain is real and reproducible on encounters never used for selection, but it is model-error compensation, not calibration, and the paper's route, selected without reference to the test set, to comparable performance is the two-stage filter instead. The two-stage filter run at the coverage band was itself declared in the pre-registration record after the consistency-selected run was seen and before any run at the declared configuration existed; both frozen runs are reported in §4.6.3 regardless of outcome, and the declaration order ships with the data record.

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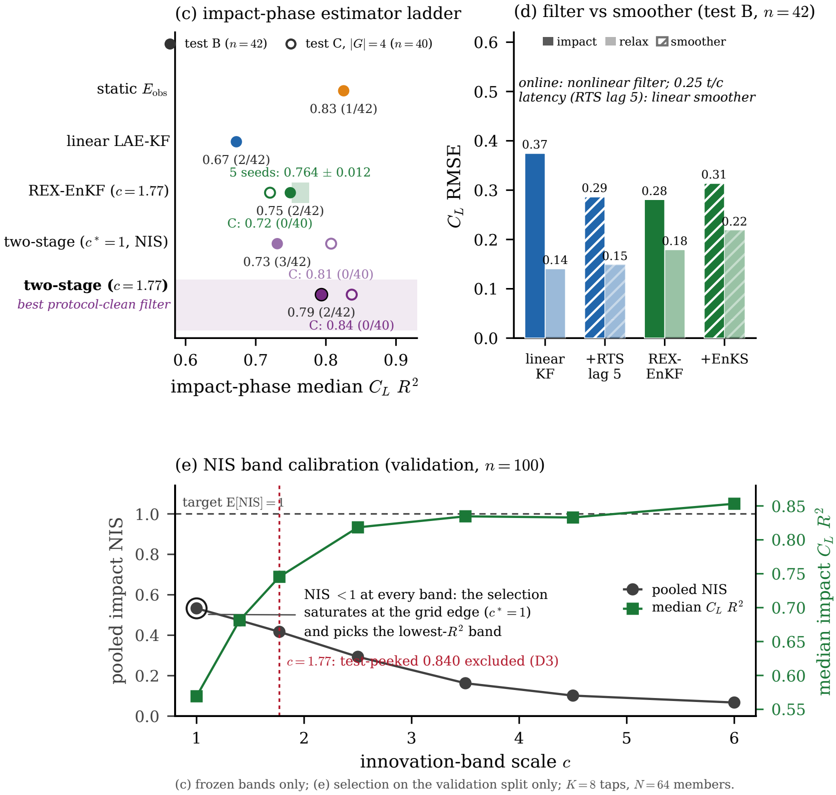

Figure 29. Estimator calibration and selection (in-distribution test and validation sets). (top left) The impact-phase estimator ladder with the five-seed member-noise band, open markers the frozen boundary test values; (top right) the filter against the fixed-lag smoother; (bottom) the band-scale calibration on the held-out training encounters, where innovation consistency and tracking disagree and the selection saturates at the grid edge.

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